

(12) **United States Patent**
Loboda et al.

(10) **Patent No.:** **US 12,410,766 B1**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **RADIAL IN-FLOW PARTICLE BED
NUCLEAR ROCKET ENGINE AND METHOD**

(71) Applicant: **Dark Fission Space Systems Inc.**,
Huntington Beach, CA (US)
(72) Inventors: **Gregory George Loboda**, Huntington
Beach, CA (US); **Fred George William
Kennedy, III**, Hillsboro, VA (US);
Michael Gordon Jacox, Magnolia, TX
(US)

(73) Assignee: **Dark Fission Space Systems Inc.**,
Huntington Beach, CA (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 54 days.

(21) Appl. No.: **18/233,554**

(22) Filed: **Aug. 14, 2023**

Related U.S. Application Data

(60) Provisional application No. 63/371,492, filed on Aug.
15, 2022.

(51) **Int. Cl.**
G21D 5/02 (2006.01)
B64G 1/40 (2006.01)
B64G 1/42 (2006.01)
F02K 9/64 (2006.01)
G21C 7/28 (2006.01)
G21C 15/06 (2006.01)
G21C 15/08 (2006.01)
G21C 15/10 (2006.01)
G21C 1/06 (2006.01)

(52) **U.S. Cl.**
CPC **F02K 9/64** (2013.01); **B64G 1/408**
(2013.01); **B64G 1/422** (2013.01); **G21C 7/28**
(2013.01); **G21C 15/06** (2013.01); **G21C**

15/08 (2013.01); **G21C 15/10** (2013.01);
G21D 5/02 (2013.01); **F05D 2260/20**
(2013.01); **G21C 1/06** (2013.01)

(58) **Field of Classification Search**
CPC .. **B64G 1/408**; **G21C 7/28**; **G21C 1/06**; **F02K**
9/42; **G21D 5/02**
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,108,054 A * 10/1963 Blackman, Jr. G21D 5/02
165/84
3,168,807 A * 2/1965 Ledwith G21D 5/02
60/233

(Continued)

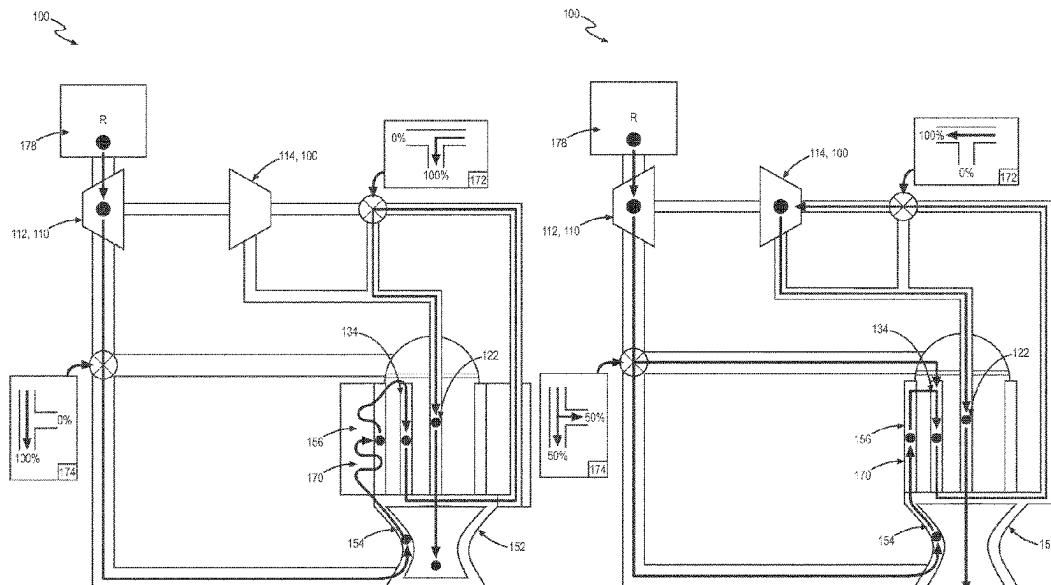
Primary Examiner — Lorne E Meade

(74) *Attorney, Agent, or Firm* — Run8 Patent Group,
LLC; Peter Miller

(57) **ABSTRACT**

A nuclear engine system includes: a pump configured to
pump a propellant; a fuel element including a set of nuclear
fuel particles; a moderator configured to surround the fuel
element and defining a set of moderator coolant channels
configured to cool the moderator; a reflector including a
neutron-reflecting material and a reflector coolant channel
arranged within the reflector to cool the reflector, the reflec-
tor configured to at a first time, operate in a closed configu-
ration to reflect neutrons to the fuel element to increase an
energy flux, and at a second time, operate in an open
configuration to leak neutrons out of the engine system to
decrease the energy flux; and a thrust nozzle configured to
outlet propellant from the fuel element to produce thrust
including a nozzle coolant channel arranged within a wall of
the thrust nozzle configured to cool the nozzle.

20 Claims, 16 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

3,286,468	A *	11/1966	Plebuch	G21D 5/02
				376/318
3,336,749	A *	8/1967	Rom	G21D 5/02
				976/DIG. 82
3,793,832	A *	2/1974	Moon	G21D 5/02
				376/385
5,475,722	A *	12/1995	Culver	G21D 5/02
				376/386
5,873,239	A *	2/1999	Culver	G21D 5/02
				376/318
2017/0263345	A1 *	9/2017	Venneri	G21C 5/02
2021/0304909	A1 *	9/2021	Gramlich	G21C 5/02
2022/0005619	A1 *	1/2022	Cisneros, Jr.	G21C 7/10
2022/0328202	A1 *	10/2022	Venneri	G21C 7/12

* cited by examiner

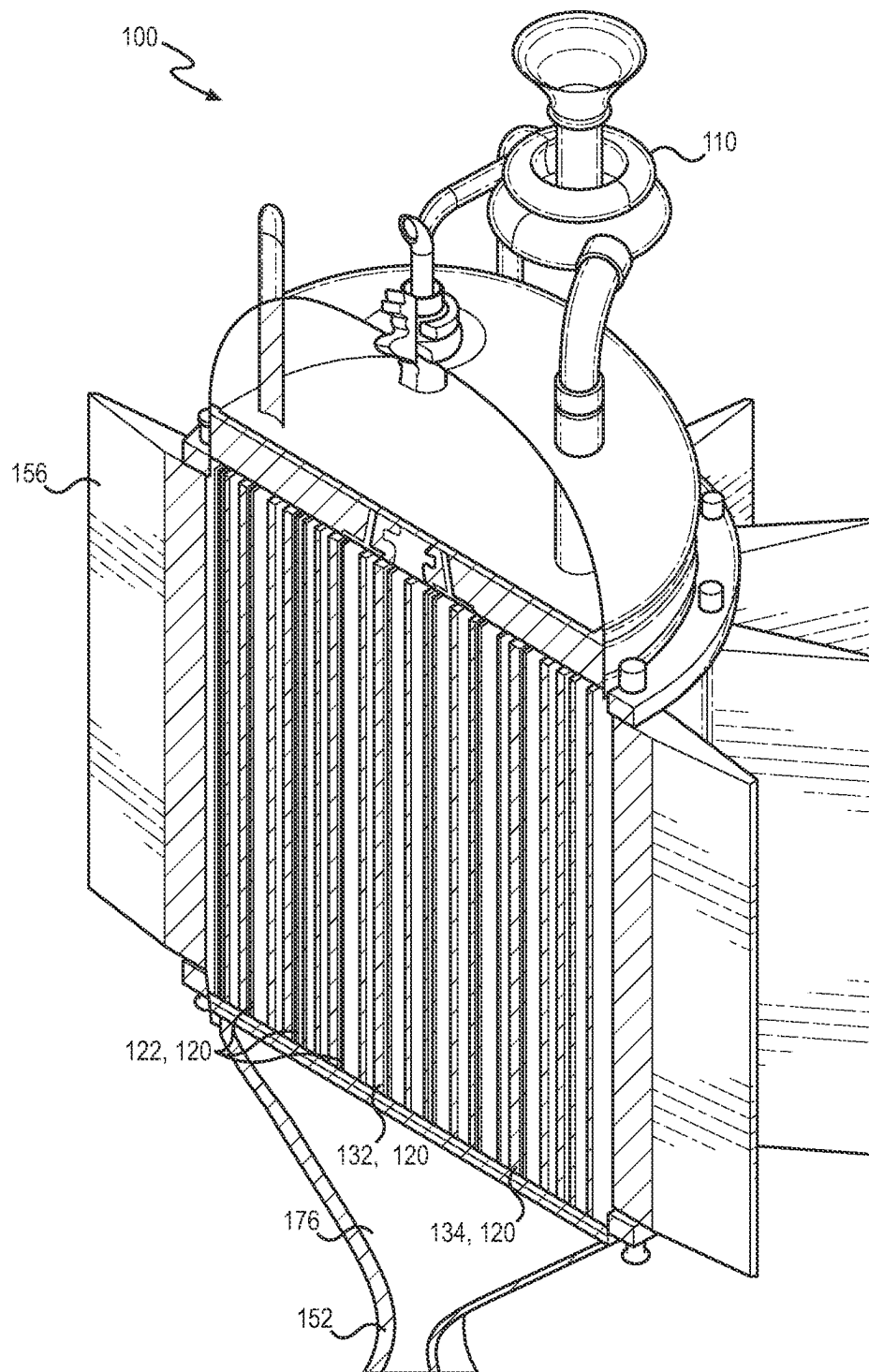


FIGURE 1

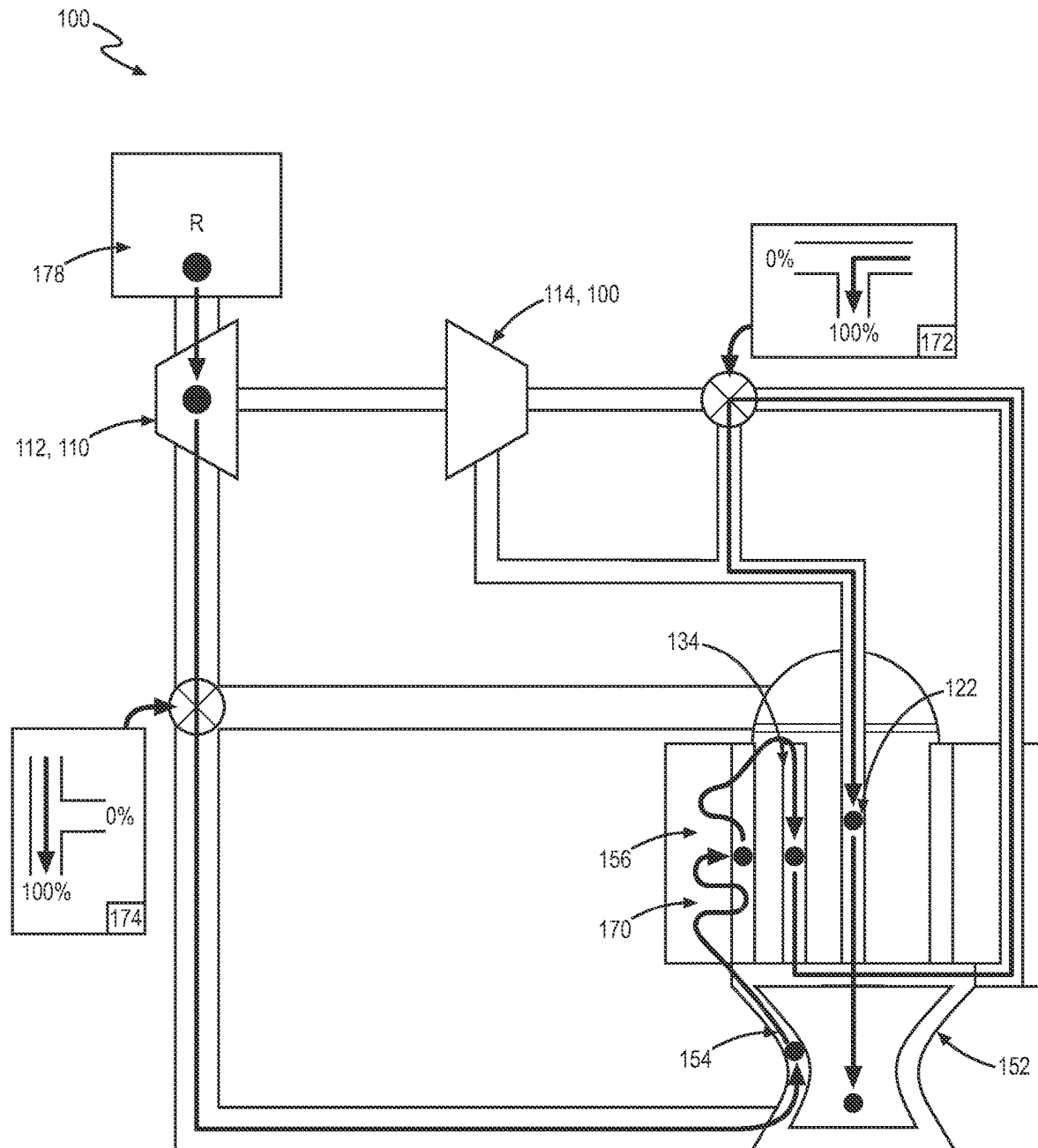


FIGURE 2A

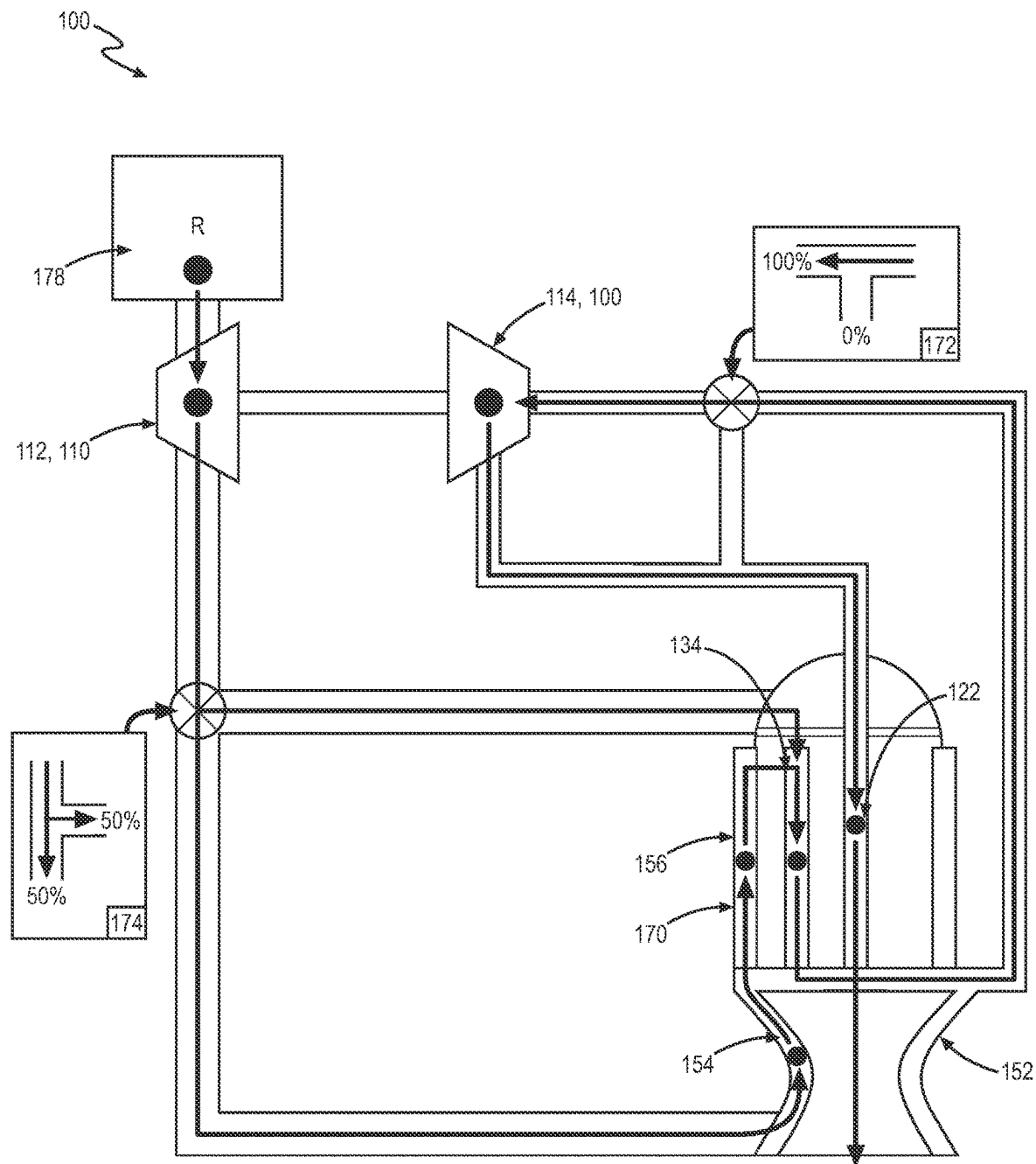


FIGURE 2B

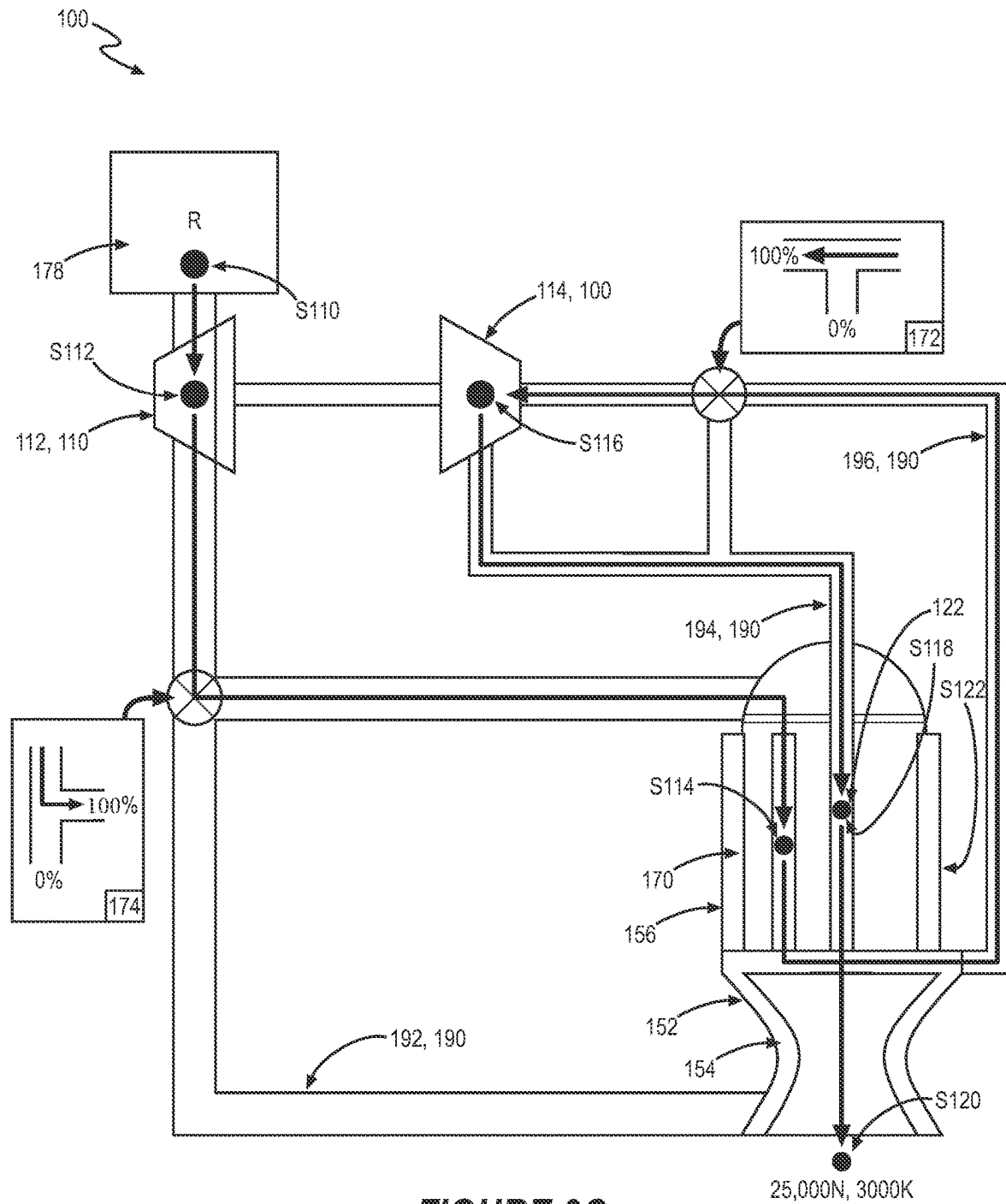


FIGURE 2C

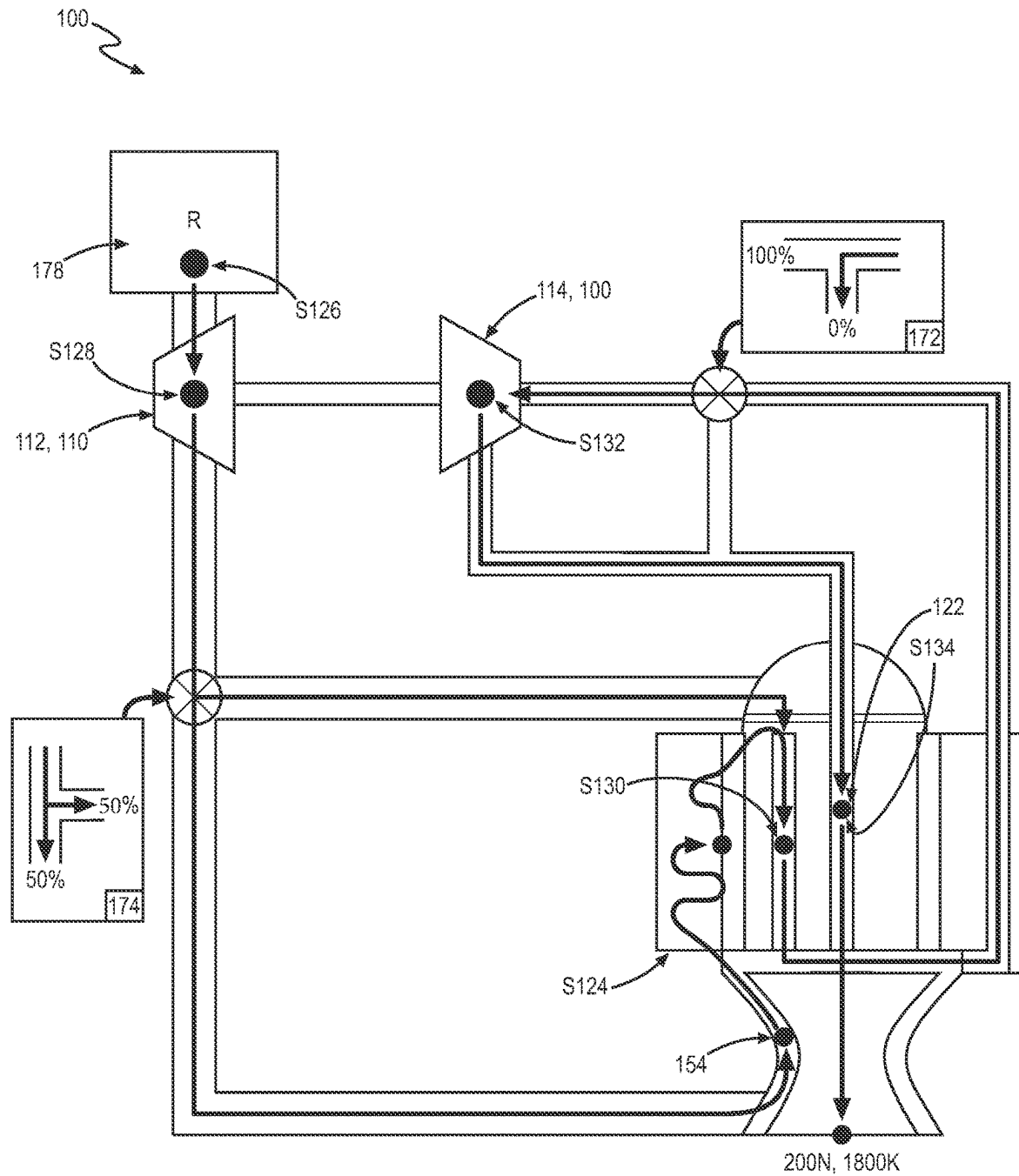


FIGURE 2D

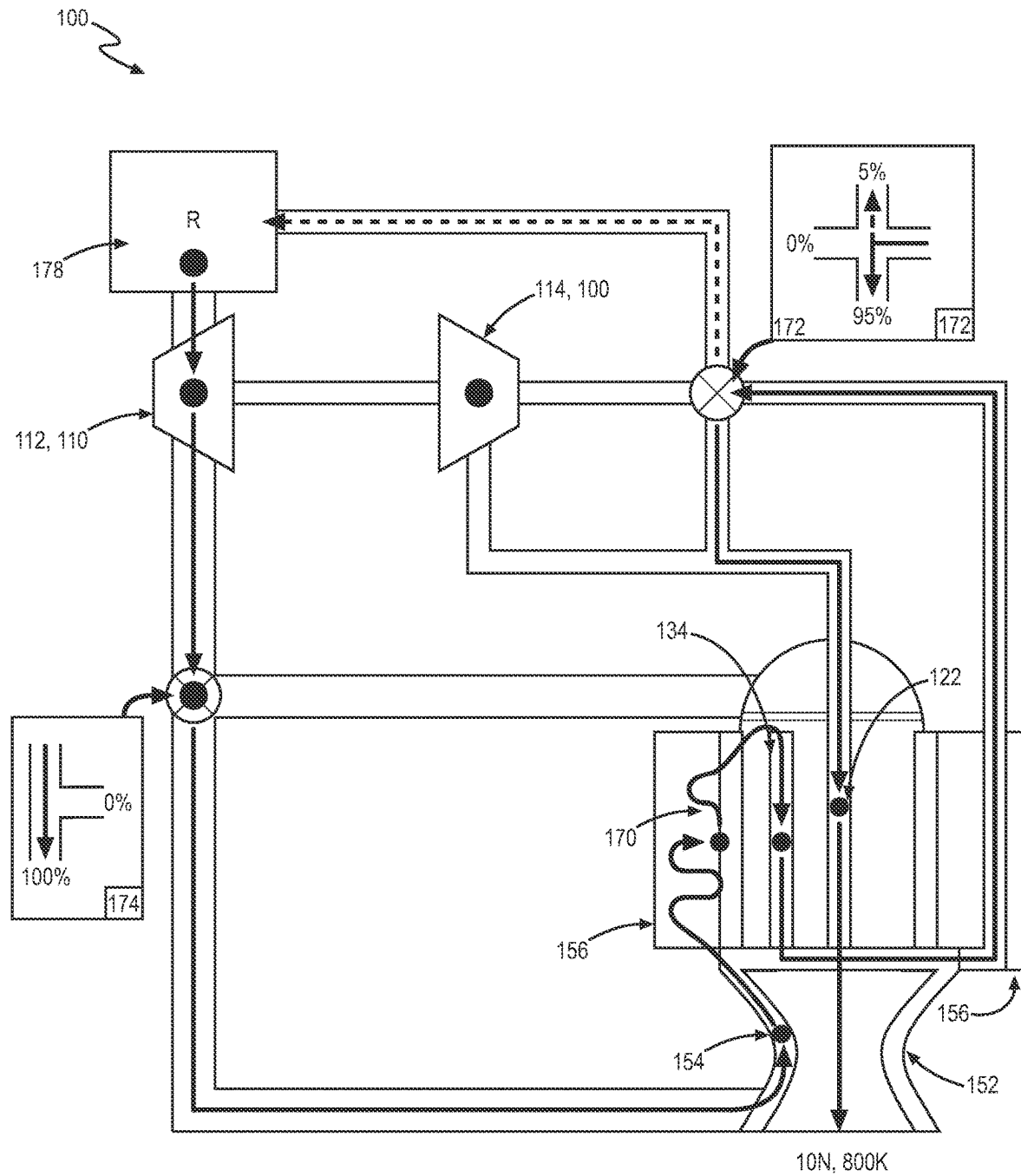


FIGURE 2E

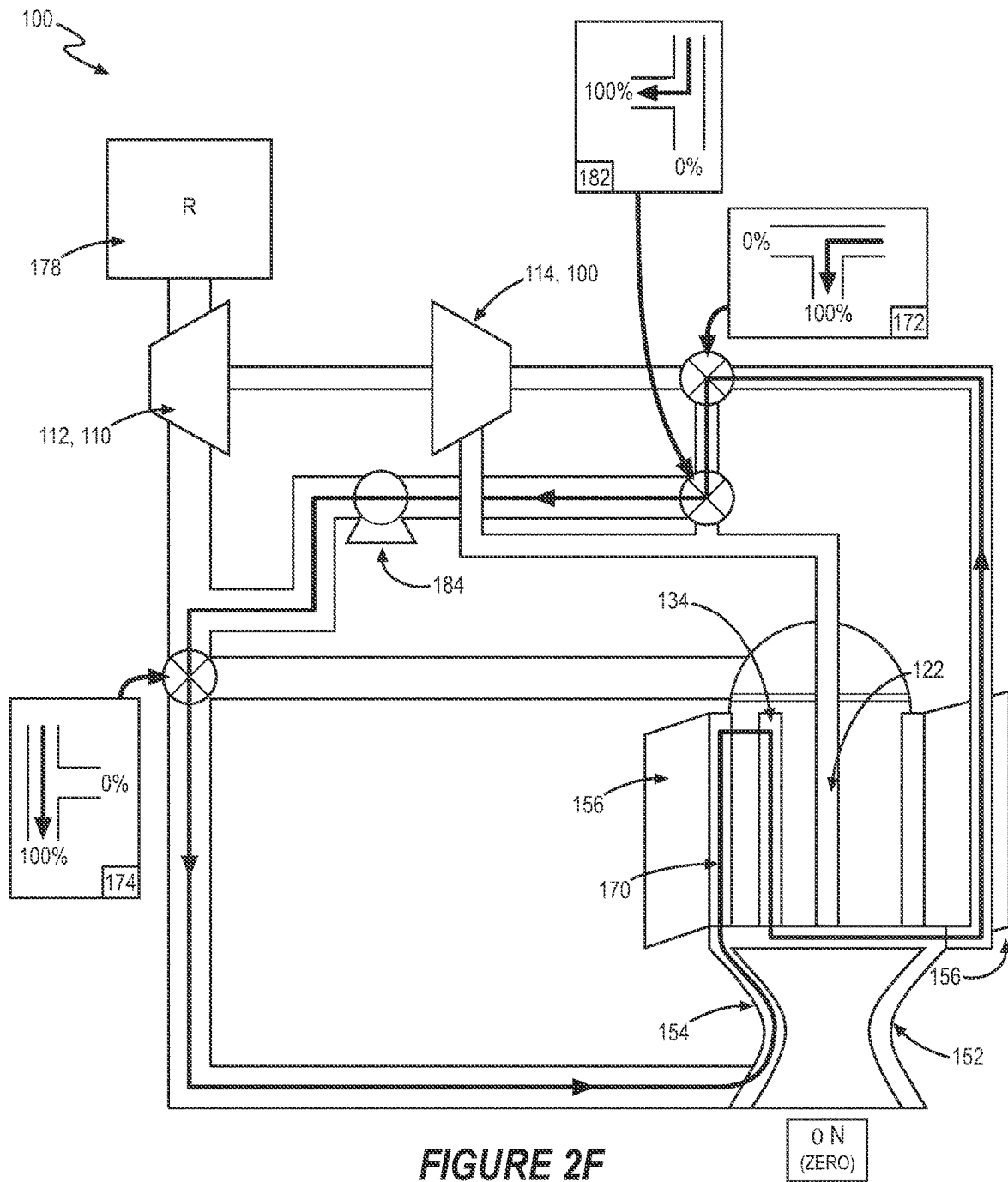


FIGURE 2F

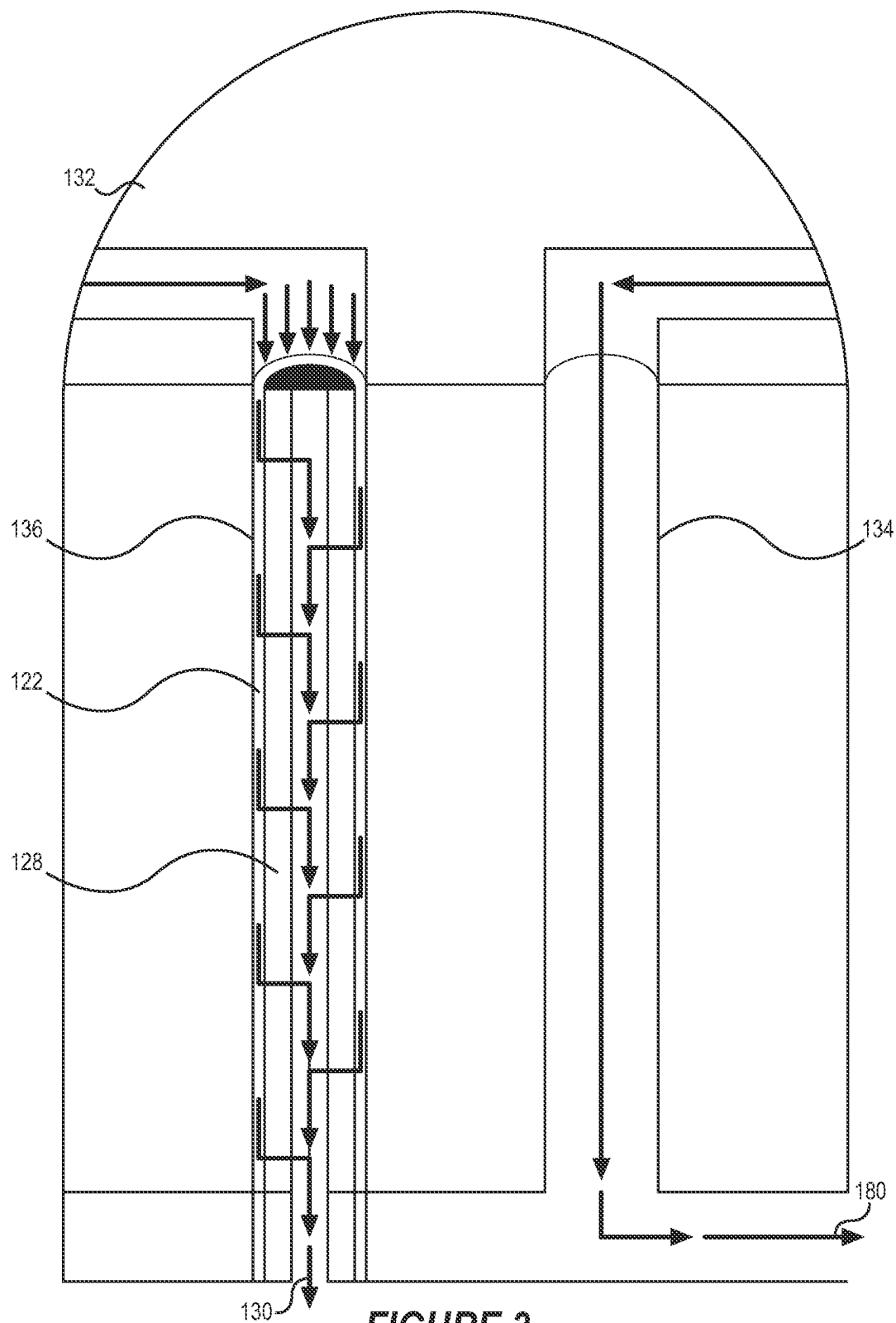


FIGURE 3

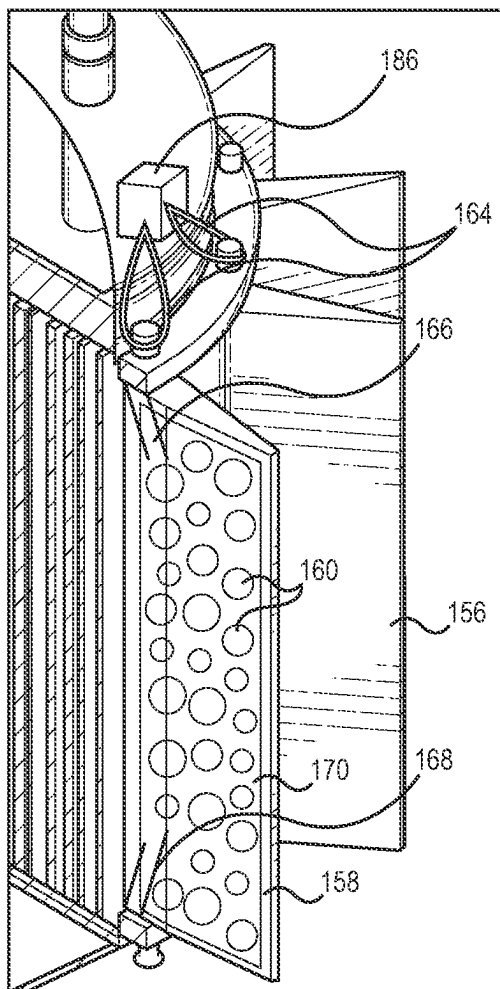


FIGURE 4A

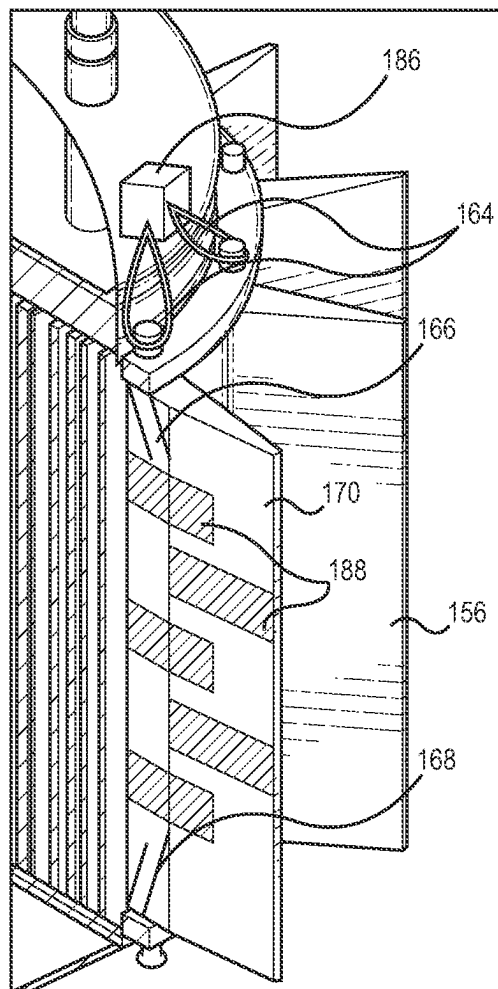
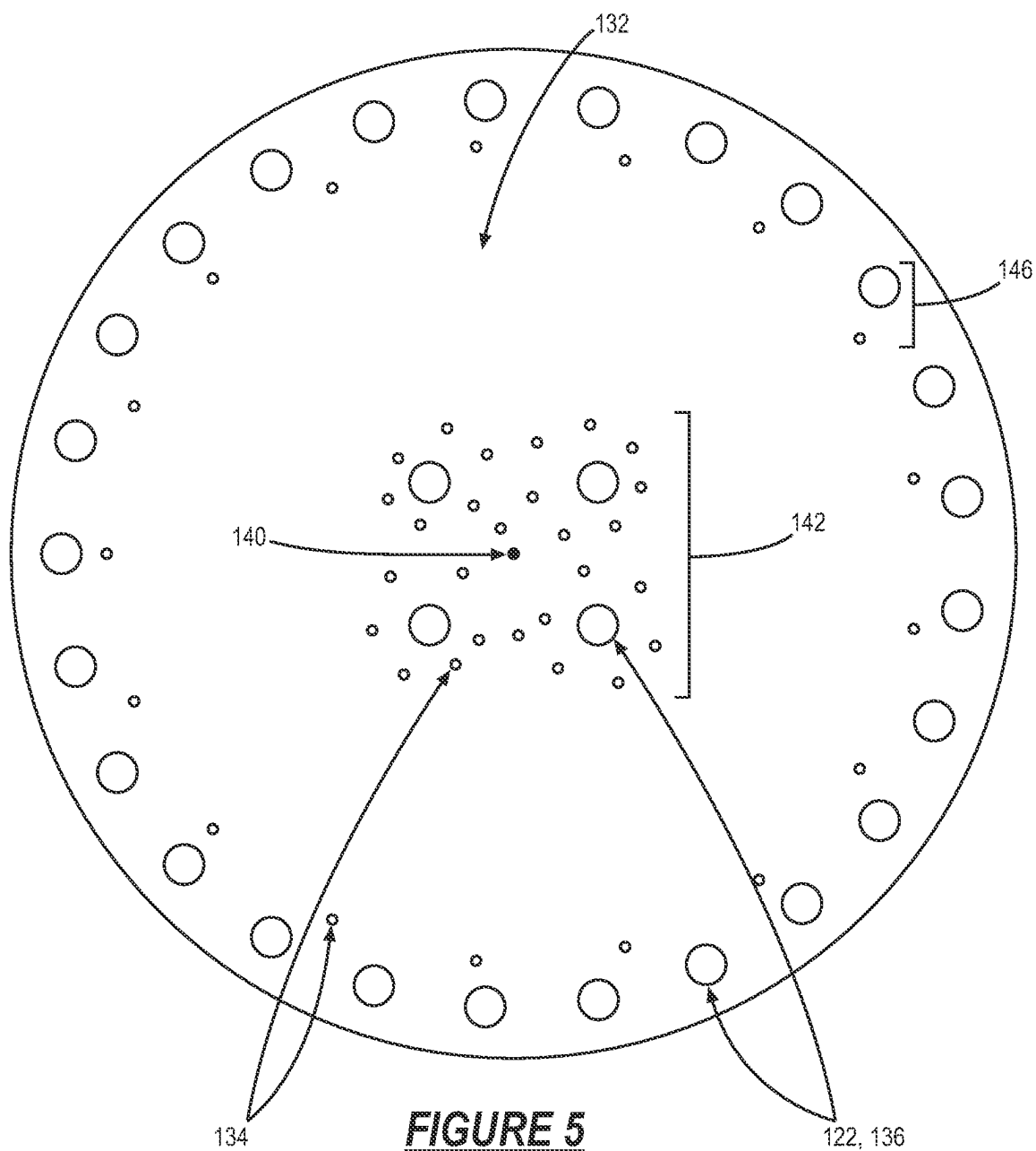
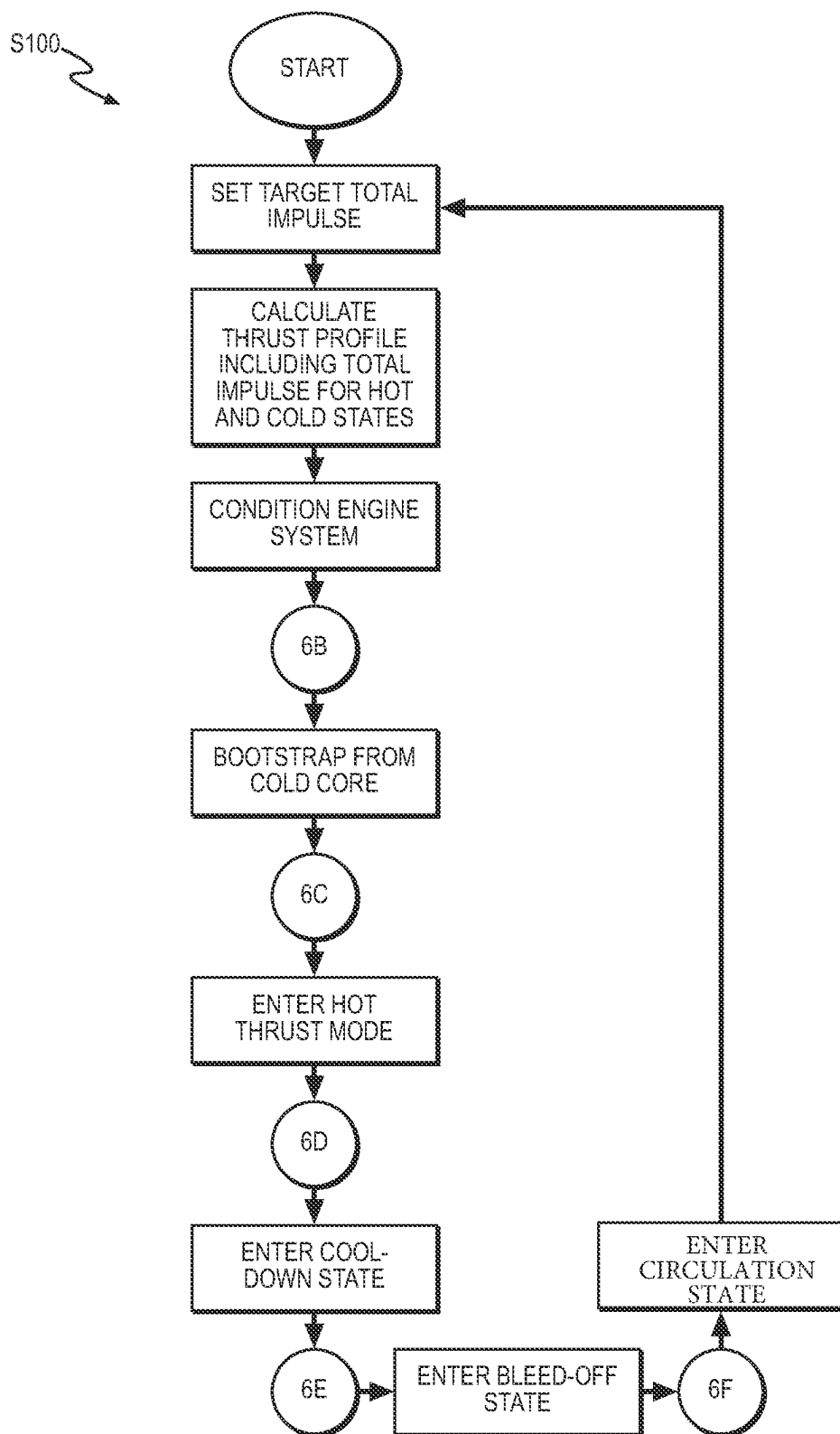
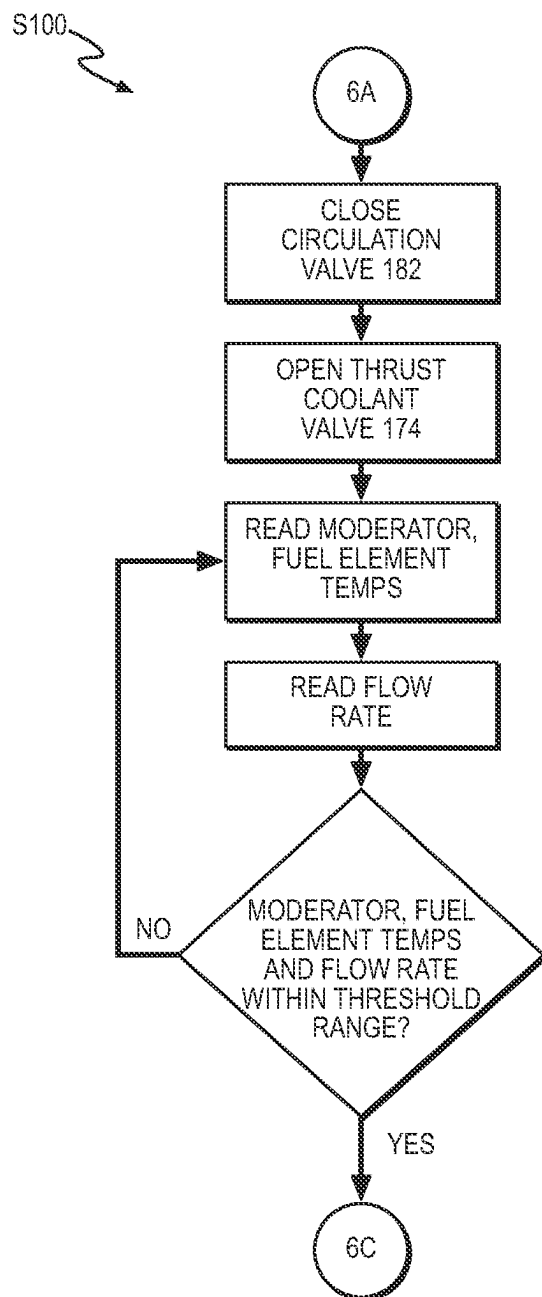
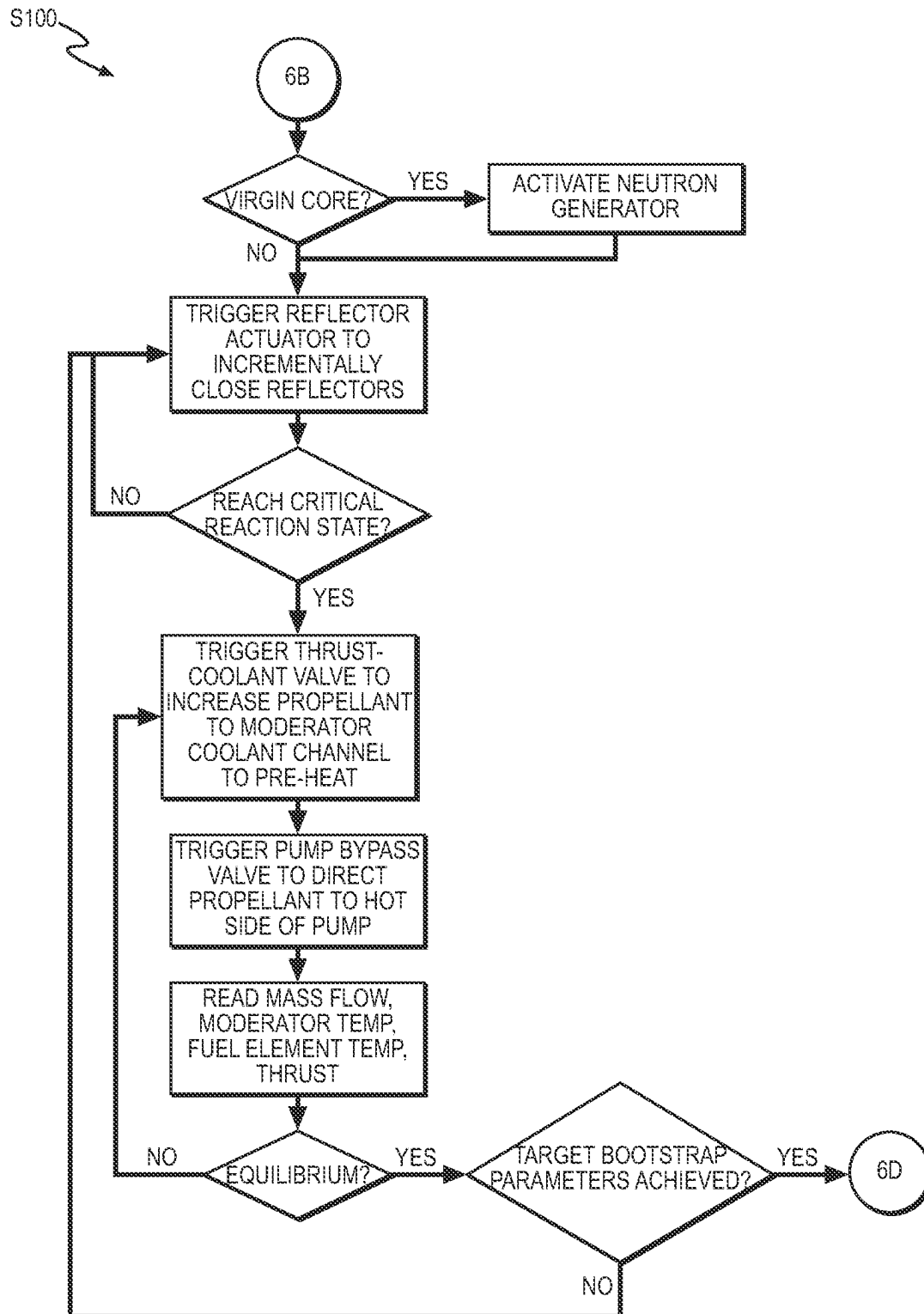


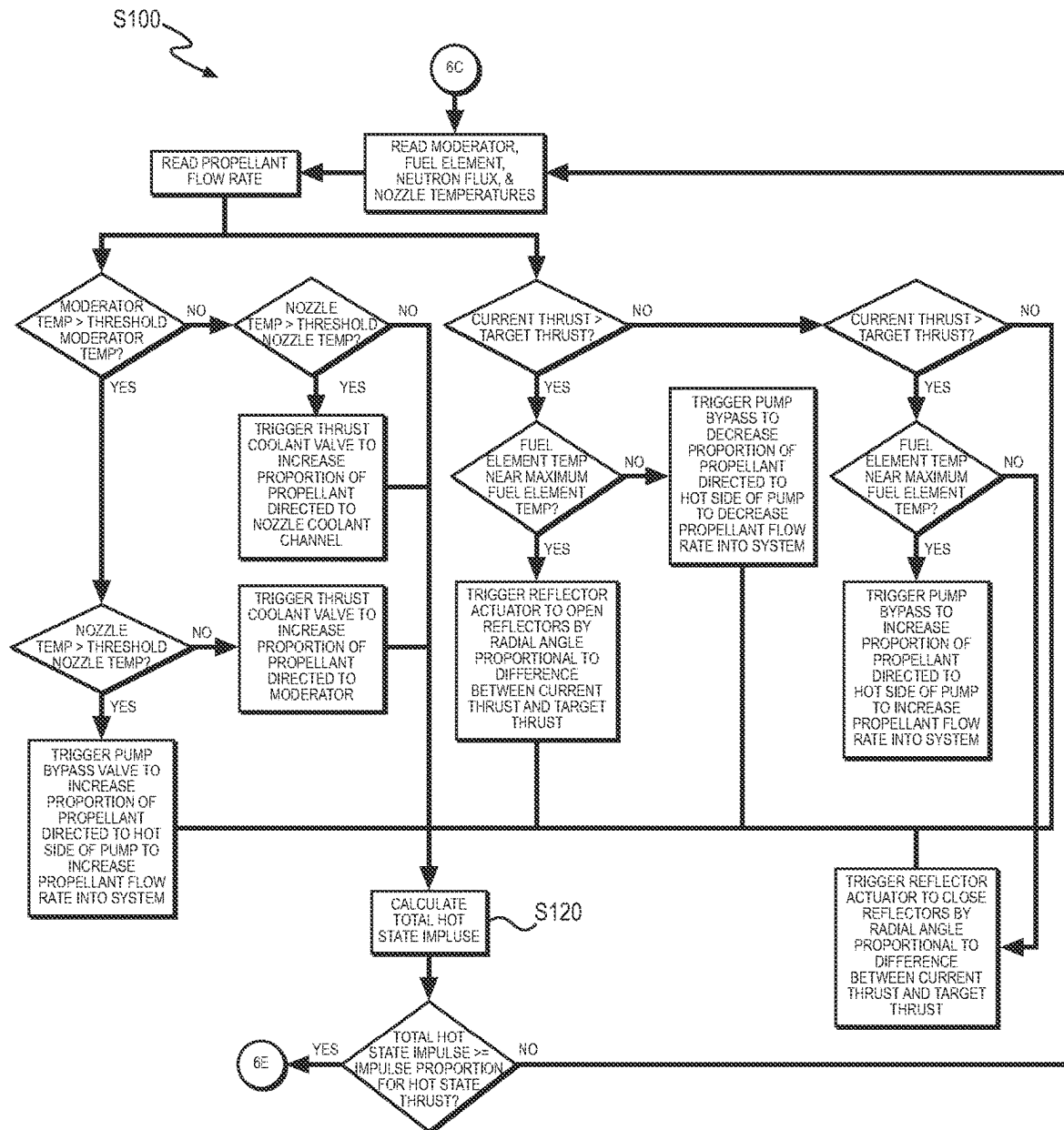
FIGURE 4B

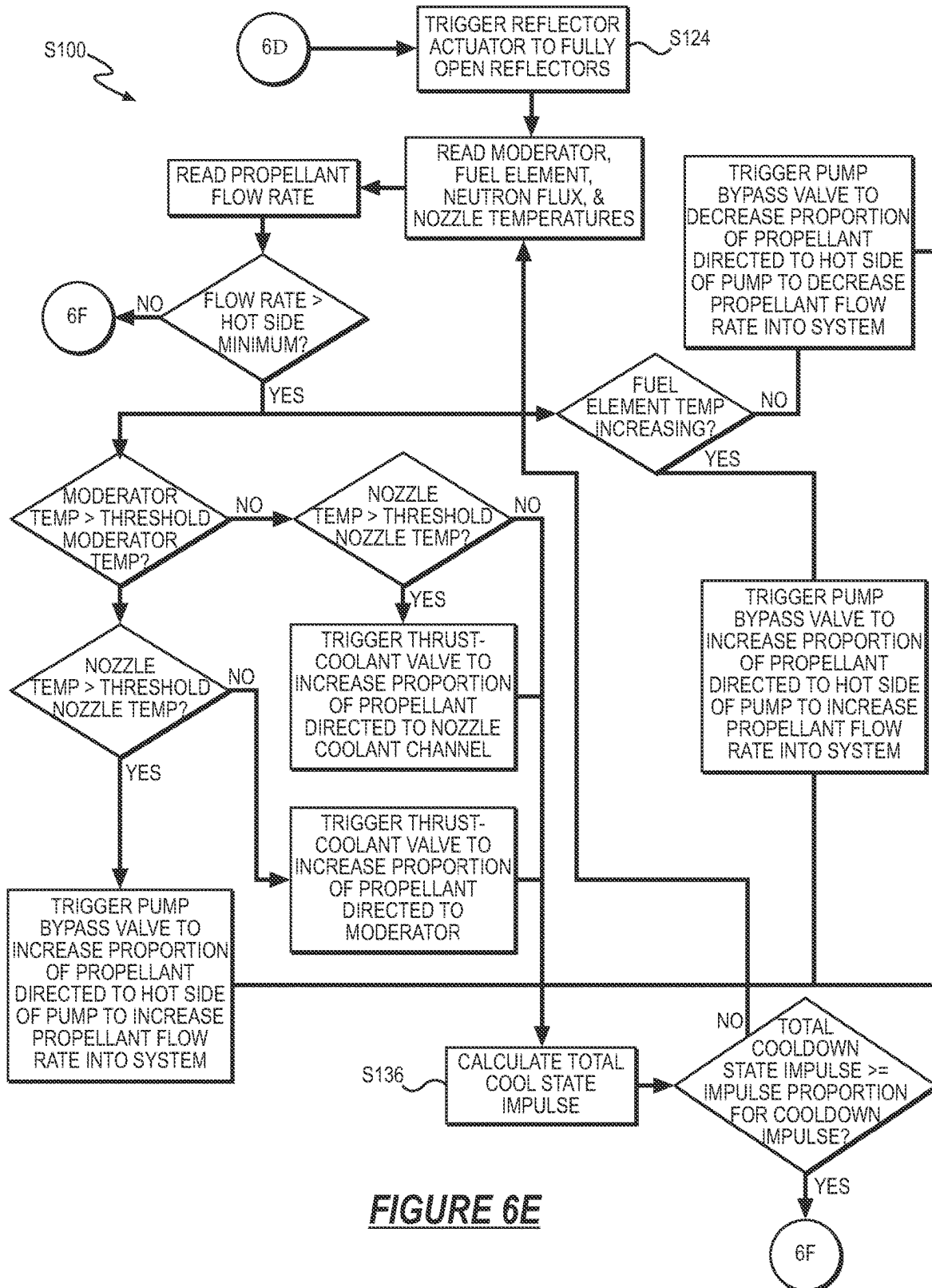


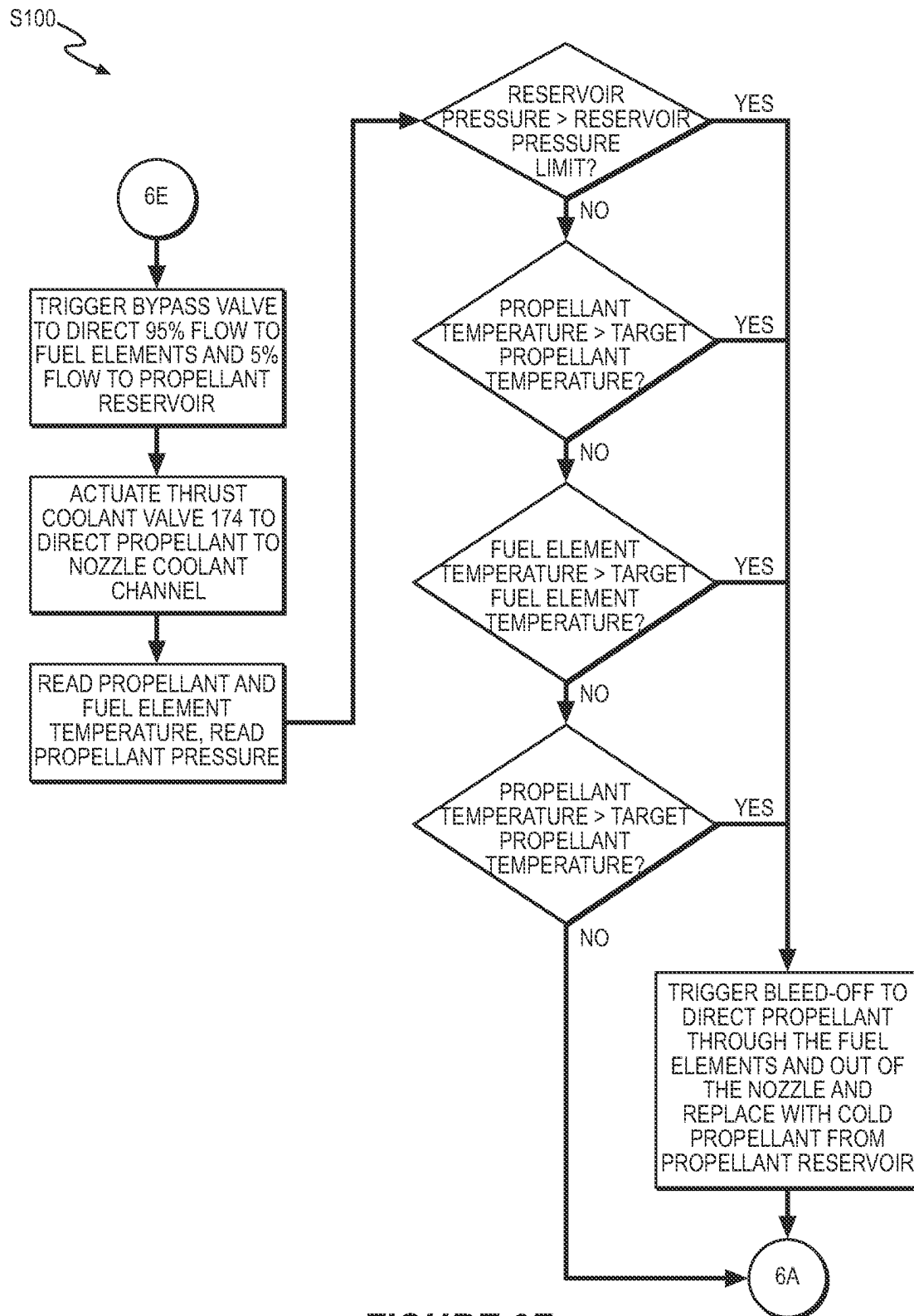
**FIGURE 6A**

**FIGURE 6B**

**FIGURE 6C**

**FIGURE 6D**



**FIGURE 6F**

1

RADIAL IN-FLOW PARTICLE BED NUCLEAR ROCKET ENGINE AND METHOD

CROSS-REFERENCE TO RELATED APPLICATIONS

This Application claims the benefit of U.S. Provisional Patent Application No. 63/371,492, filed on 15 Aug. 2022, which is incorporated in its entirety by this reference.

TECHNICAL FIELD

This invention relates generally to the field of nuclear thermal rocket propulsion and more specifically to a new and useful radial in-flow particle fuel bed reactor and an associated engine cycle of a nuclear thermal rocket propulsion system.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 is a representation of a rocket including one variation of a nuclear engine system;

FIGS. 2A, 2B, 2C, 2D, 2E, and 2F are a schematic representations of one variation of a method;

FIG. 3 is a representation of one variation of the nuclear engine system;

FIGS. 4A and 4B are schematic representations of variation of the nuclear engine system;

FIG. 5 is a representation of one variation of the nuclear engine system; and

FIGS. 6A, 6B, 6C, 6D, 6E, and 6F are flowcharts representing one variation of the method.

DESCRIPTION OF THE EMBODIMENTS

The following description of embodiments of the invention is not intended to limit the invention to these embodiments but rather to enable a person skilled in the art to make and use this invention. Variations, configurations, implementations, example implementations, and examples described herein are optional and are not exclusive to the variations, configurations, implementations, example implementations, and examples they describe. The invention described herein can include any and all permutations of these variations, configurations, implementations, example implementations, and examples.

1. SYSTEM

As shown in FIG. 1, an engine system **100** includes: a pump; a reactor assembly; a reflector; a thrust chamber; and an exhaust nozzle.

The pump **110** is configured to pump propellant throughout the engine system **100**.

The reactor assembly **120** includes a set of fuel elements **122** and a moderator **132**. Each fuel element **122** in the set of fuel elements **122** includes a set of low enriched uranium fuel particles defining a set of interstitial volumes through which the pump **110** pumps the propellant.

The moderator **132** defines a set of fuel bores, each fuel bore: fluidly coupled to a warm propellant inflow manifold; occupied by a fuel element; and configured to direct warm propellant axially downward from the warm propellant inflow manifold along a length of the fuel element. The moderator **132** additionally defines a set of moderator coolant channels, each moderator coolant channel: fluidly isolated from the set of fuel bores; defining a set of moderator

2

coolant inlets fluidly coupled to the thrust-coolant valve; defining a set of moderator coolant outlets fluidly coupled to the pump bypass valve; and configured to pass propellant through the moderator to cool the moderator.

The reflector **156** is arranged on a perimeter of the reactor assembly **120** and includes: a neutron-reflecting material; and a reflector coolant channel arranged within the reflector configured to exchange thermal energy with propellant pumped through the reflector coolant channel. The reflector **156** is operable in: a first position, the reflector defining a first cross-section of the neutron-reflecting material facing the fuel elements in the first position and reflecting incident neutrons toward the fuel elements at a first rate; and a second position, the reflector defining a second cross-section of the neutron-reflecting material facing the fuel elements in the second position and reflecting incident neutrons toward the fuel elements at a second rate, the second cross-section less than the first cross-section, the second rate less than the first rate.

The thrust nozzle **152**: is configured to accelerate and expel propellant to produce thrust; and includes a nozzle coolant channel arranged within a wall of the thrust nozzle configured to exchange thermal energy with propellant pumped through the nozzle coolant channel.

1.1 Variation: Fuel Element Geometry

In one variation, each fuel element defines a cylindrical geometry of concentric layers including: a cold shell defining a first array of perforations configured to direct the propellant radially through the cold shell; a fuel bed containing the set of low enriched uranium fuel particles; a hot shell defining a second array of perforations configured to direct the propellant radially through the hot shell; and an interior chamber configured to direct propellant axially out of the fuel element toward the nozzle.

1.1.1 Variation: Reflector

In another variation, the reflector **156** of the engine system **100** is operable in a closed position and in an open position (and in intermediate positions between the closed and open configurations).

In the closed position, the reflector **156** occupies a shallow angle (e.g., less than 5°) relative to a tangent of the perimeter of the reactor assembly **120** such that a widest cross-section of a neutron-reflective region of the reflector (e.g., a solid beryllium surface; a matrix of beryllium oxide pellets) faces the set of fuel elements **122** and reflects a first proportion of neutrons—emitted by the set of nuclear fuel particles within the set of fuel elements **122** as a product of nuclear fission—back into the set of fuel elements to increase a frequency of neutron-neutron collisions within the reactor assembly **120** and thus increase a rate of fission within the fuel elements.

In the open position, the reflector **156**: occupies a wide angle (e.g., approximately 90°) relative to the tangent of the perimeter of the reactor assembly **120** such that a narrowest cross-section of the neutron-reflective region of the reflector faces the set of fuel elements **122**; and reflects a second proportion of neutrons—much less (e.g., 80% less) than the first proportion of neutrons—back into the set of fuel elements to reduce a frequency of neutron-neutron collisions within the set of fuel elements **122** and thus reduce a rate of fission within the reactor assembly to subcritical levels.

1.2 Applications

Generally, an engine system **100** within a nuclear rocket includes: a pump; a set of fuel elements **122**; a moderator

132; a neutron reflector; and a thrust nozzle. The pump 110 pumps propellant through the engine system 100 to: cool components of the engine system 100 with low temperature propellant; heat propellant to an exhaust temperature by directing it through the set of fuel elements; and produce thrust by expelling high temperature propellant through the thrust nozzle.

The engine system includes reactor assembly 120 including a set of fuel elements 122 arranged within a moderator 132. Each fuel element 122 includes: a cold shell 124; a fuel bed 128; and a hot shell 126. The cold shell 124 and the hot shell 126 each include perforations configured to direct a flow of propellant radially inward from the moderator 132, through the cold shell 124, through the fuel bed 128, and through the hot shell 126 to a fuel element outlet 130. The fuel bed 128 includes a set of nuclear fuel particles that define a set of interstitial volumes through which the propellant flows. As propellant flows through the interstitial volumes of the fuel bed 128, the propellant absorbs thermal energy from the fission reaction occurring at the nuclear fuel particles. The propellant thereby cools the nuclear fuel particles within the fuel bed 128 and absorbs thermal energy to produce thrust. The radial flow of propellant into each fuel element 122 allows the engine system 100 to heat a large volume of propellant from as low as 30K to as high as 3000K over a flow path of one to four centimeters.

The nuclear rocket is configured to complete hundreds of trips through space, such as to ferry a payload (e.g., supplies for space missions) between celestial bodies including planets, moons, and space stations. To produce thrust, the components of the engine system 100 can reach temperatures of 3000K. Therefore, the engine system 100 is configured to withstand hundreds of extreme temperature cycles ranging from the inlet propellant temperature (e.g., 30K) to the critical reaction temperature within the fuel bed 128 of each fuel element 122 (e.g., 3000K). As shown in FIG. 3, the radial inflow configuration of the fuel elements allows for the moderator 132 to contact only low temperature (e.g., below 400K) propellant and remain isolated from the high temperature within the axial center 140 of the fuel element 122. As the propellant flows radially inward through the fuel bed 128, the propellant increases to a temperature of up to 3000K at the axial center 140 of the fuel element 122.

The engine system 100 can include an array of reflectors 156 configured to reflect neutrons into and release neutrons from the engine system to increase or decrease the rate of fission within the fuel elements. The reflectors 156 modulate the power output and operating duration of the nuclear rocket engine system 100 by moderating the rate at which neutron-neutron collisions occur within the moderator and set of fuel elements. For example, the reflectors 156 can actuate via an actuator to a closed position to reflect neutrons into the set of fuel elements and increase the incidence of neutron-neutron collisions, thereby increasing the fission rate, the temperature of the fuel elements, and the thrust produced. The reflectors 156 can also actuate to an open position to release neutrons from the moderator to decrease the incidence of neutron-neutron collisions, thereby decreasing the fission rate, the temperature of the fuel elements, and the thrust produced. The reflectors 156 thereby moderate a use rate of the fuel to increase the rocket lifetime.

Additionally, the engine system 100 defines several fluid circuits configured to distribute propellant between components of the engine system. For example, the moderator 132 includes a set of coolant channels thermally isolated from the set of fuel elements. The engine system 100 can direct propellant from the propellant reservoir 178 directly to the

moderator coolant channels 134 to maintain the temperature of the moderator 132 below the maximum operating temperature. The thrust nozzle 152 and reflectors 156 also include coolant channels. The thrust nozzle 152 can absorb heat from the propellant exiting the thrust nozzle 152 outlet. The thrust nozzle 152 includes a coolant channel within a wall of the thrust nozzle 152 in order to direct propellant from the propellant reservoir 178 to the nozzle coolant channel 154 to cool the thrust nozzle. Similarly, the reflector coolant channel 170 can receive propellant from the propellant reservoir 178 to cool the reflector, to maintain the reflector 156 under a threshold temperature.

1.3 Engine System Structure

Generally, the engine system 100 is a portion of a rocket that produces thrust to propel the rocket through space. In one implementation, thrust represents an instantaneous measure of force output by the rocket at a given time while impulse defines a total amount of force produced by the rocket over a period of time. Therefore, impulse defines thrust integrated over a period of time. As described within, the engine system 100 can: produce and measure an instantaneous thrust; and produce and measure impulse over time.

Typically, the rocket includes: a payload; a propellant reservoir 178; and an engine system 100. The payload is arranged at a first end of the rocket opposite a second end of the rocket including the engine system 100. The payload can include: a cargo hold configured to store supplies transported by the rocket; and a set of electronics. The propellant reservoir 178 is: arranged upstream of the pump; interposed between the set of fuel elements and a payload; and configured to store a volume of the propellant including hydrogen, the volume of the propellant absorbing neutrons emitted axially by the set of fuel elements during a fission reaction to shield the payload from radiation.

The engine system 100 is arranged at the second end of the rocket and fluidly coupled to the propellant reservoir 178. The engine system 100: receives propellant from the propellant reservoir 178; heats the propellant; and releases the heated propellant to produce thrust to propel the rocket. Generally, the engine system 100 includes: a set of fuel elements 122; a moderator 132; a thrust nozzle; a reflector; a pump; and a fluid circuit (e.g., including pipes and/or valves) through which propellant flows between each of the components.

In one implementation, the reactor assembly defines a particle bed reactor. However, the engine system can alternatively be configured with other nuclear reactor and nuclear fuel types.

1.4 Set of Fuel Elements

Generally, the engine system includes a set of fuel elements. Each fuel element 122 includes a fuel bed 128 containing a set of nuclear fuel particles (e.g., low-enriched uranium kernels or "LEU", high assay low enriched uranium or "HALEU", high enriched uranium or "HEU") configured to release thermal energy via nuclear fission. Each fuel element 122 is configured for propellant to flow through the fuel bed 128 to: cool the nuclear fuel particles; and heat the propellant by absorbing thermal energy from the nuclear fission reactions of the nuclear fuel particles. Each fuel element 122 defines an outlet configured to direct heated propellant from the fuel element 122 to a thrust chamber 176 and/or a thrust nozzle 152 of the engine system 100.

1.4.1 Fuel Element Arrangement

In one implementation, the reactor assembly **120** includes the set of fuel elements **122** arranged within the moderator at a uniform density with a common pitch distance.

In another implementation, the set of fuel elements are arranged in a non-uniform density within the moderator, including: a first cluster of fuel elements arranged in a low-density distribution about the center of the moderator and offset by a first pitch distance between axial centers of these fuel elements; a second cluster of fuel elements arranged in a moderate-density distribution about the first cluster of fuel elements and offset by a second pitch distance—between axial centers of these fuel elements—less than the first pitch distance; and a third cluster of fuel elements arranged in a high-density distribution about the second cluster of fuel elements and offset by a third pitch distance—between axial centers of these fuel elements—less than the second pitch distance. Generally, at any instant in time, a highest density of free neutrons may be present proximal a fuel element located near the center of the moderator—given a nominal fission reaction rate—due to presence of other reactive fuel elements filling a large proportion of the radial field of view of this fuel element. Conversely, at any instant in time, a lowest density of free neutrons may be present proximal a fuel element located near the perimeter of the moderator—given the nominal fission reaction rate—due to the absence of other reactive fuel elements in a large proportion of the radial field of view of this fuel element. Therefore, the system can include a lower density of fuel elements near the center of the moderator and a higher density of fuel elements near the perimeter of the moderator to achieve more uniform availability of neutrons to fuel elements throughout the moderator, thereby achieving more uniform fission rates and propellant output temperatures across these fuel elements.

In one implementation, the set of fuel elements are arranged in clusters, as shown in FIG. 5, to produce an approximately uniform radial and longitudinal thermal power density within the reactor assembly **120**. In one example, the set of fuel elements includes: a first cluster **142** of fuel elements; and a second cluster **146** of fuel elements. In this implementation, the first cluster **142** is arranged in a first region of the moderator distal a reflector **156** that defines an opening of the engine system **100** to the external environment. The first cluster **142** is characterized by a first offset distance between each fuel element **122**. The second cluster **146** is arranged in a second region of the moderator proximal a reflector **156** defining an opening to the external environment. The second cluster **146** is characterized by a second offset distance less than the first offset distance between each fuel element **122** of the second cluster **146**. The fuel elements of the first cluster **142** exhibit a higher neutron availability than the fuel elements of the second cluster **146**. Therefore, the fuel elements of the first cluster **142** define an offset distance higher than the offset distance of the second cluster **146** to balance the thermal power output of the first cluster **142** and the second cluster **146**.

1.4.2 Fuel Element Structure

In one implementation, a fuel element **122** of the reactor assembly **120** includes: a cold shell **124**; a compliance structure; a fuel bed **128**; and a hot shell **126**. The cold shell **124**: defines a cylindrical geometry including an array of cold perforations and an interior cold surface; is arranged within a moderator bore **136** of a moderator **132**; and is configured to direct propellant radially inwardly from the moderator bore **136** through the array of cold perforations, and toward the fuel bed **128**.

The fuel bed **128** includes a set of nuclear fuel particles and is configured to release thermal energy via nuclear fission. The set of nuclear fuel particles defines a set of interstitial volumes through which propellant can flow. In one implementation, the fuel bed **128** includes a set of approximately spherical, 1 mm diameter, low-enriched uranium kernels with a 55-70% packing density.

The compliance structure: is interposed between the fuel bed **128** and the cold shell **124**; isolates the interior cold surface from direct contact with the fuel bed **128** over a range of operating temperatures of the fuel bed **128**; is configured to pass propellant from the cold shell **124** toward the fuel bed **128**; and is configured to elastically deform radially and longitudinally to absorb thermal movement of the fuel bed **128** over the range of operating temperatures.

The hot shell **126**: defines a cylindrical geometry including an array of hot perforations, an interior chamber, and an outlet port; is coaxial with the cold shell **124**; cooperates with the compliance structure to contain the fuel bed **128**; and is configured to direct propellant, heated by the fuel bed **128**, radially inwardly from the fuel bed **128** into the interior collection volume. The interior collection volume is configured to collect propellant longitudinally out of the fuel element **122** via the outlet port to the thrust chamber **176** and/or thrust nozzle.

1.5 Moderator

The engine system **100** includes a moderator **132** that defines a first set of fuel bores, each fuel bore: fluidly coupled to a warm propellant inflow manifold; occupied by a fuel element; and configured to feed warm propellant axially downward from the warm propellant inflow manifold along a height of the fuel element. The moderator defines a second set of coolant channels: fluidly isolated from the first set of fuel bores; defining moderator coolant inlets fluidly coupled to the thrust-coolant valve; defining moderator coolant outlets fluidly coupled to the pump bypass valve; and configured to pass propellant through the moderator to cool the moderator and cold shells of the fuel elements.

In one implementation, the moderator **132** is characterized by a hydrogenated homogeneous or heterogeneous material. For example, the moderator **132** can include a polymer material (e.g., ultra-high-molecular-weight polyethylene, or “UHMWPE,” “UHMW”) and occupies a solid state below the maximum operating temperature. In another example the moderator can include metal hydrides (e.g., LiH, ZrH, YH) or beryllium-based materials (e.g., Be, and BeO). In this implementation the moderator **132** includes: a set of bores **136** configured to support each fuel element **122** of the set of fuel elements; and the set of moderator coolant channels **134** isolated from the set of bores **136** wherein the set of moderator coolant channels **134** receive propellant to maintain a temperature of the moderator **132** below the maximum operating temperature. Further, each bore **136** of the moderator **132** can define a fluted or splined geometry configured to: support each fuel element **122** within the moderator **132** and define propellant channels through which propellant can flow from the moderator **132** through the perforations of the cold shell **124**.

In another implementation, the moderator coolant channels **134** are arranged in clusters including: a first cluster of moderator coolant channels arranged in a high-density distribution about the center of the moderator and offset by a first pitch distance between axial centers of each moderator coolant channel; a second cluster of moderator coolant channels arranged in a moderate-density distribution about

the first cluster of moderator coolant channels and offset by a second pitch distance—between axial centers of each moderator coolant channel—greater than the first pitch distance; and a third cluster of moderator coolant channels arranged in a low-density distribution about the second cluster of moderator coolant channels and offset by a third pitch distance—between axial centers of each moderator coolant channel—greater than the second pitch distance. In this way, the moderator coolant channel **134** are arranged based on the thermal exchange rate of the region in which they are located within the engine system **100**.

In one implementation, the moderator defines; a first inlet coupled to the set of fuel elements configured to direct warm propellant to the set of fuel elements; and a second inlet coupled to the set of moderator coolant channels configured to direct cool propellant (e.g., from the cold side of the pump) to the set of moderator coolant channels. The first inlet and the second inlet are fluidly isolated such that the first inlet and the set of moderator bores form a first fluid circuit, and the second inlet and the set of moderator coolant channels form a second fluid circuit wherein the first fluid circuit is fluidly sealed from the second fluid circuit to prevent mixing of the two propellant flows through the two fluid circuits.

The moderator defines a moderator coolant outlet configured to outlet the propellant from the moderator coolant channels and direct the propellant out of the moderator. The moderator coolant outlet is fluidly isolated from the set of moderator bores containing the set of fuel elements. Further, each fuel element defines an internal chamber configured to expel heated propellant axially to an end of the fuel element opposite the first inlet of the moderator. Each fuel element expels heated propellant away from the moderator and to the thrust nozzle. The internal chamber of each fuel element and the thrust nozzle are fluidly isolated from the moderator coolant outlet in order to prevent mixing of the flows of propellant.

1.6 Example Set of Fuel Elements Construction

In one implementation, the moderator **132** defines an approximately cylindrical geometry including: the set of bores **136** arranged parallel the longitudinal axis of the moderator **132**; and the set of moderator coolant channels **134** arranged parallel to the longitudinal axis of the moderator **132** and offset from the fuel element **122** bores **136**. The set of fuel elements extends within the bores **136** of the moderator **132**. In this implementation, the reflector **156** or array of reflectors **156** are arranged along a perimeter of the cylindrical geometry of the moderator **132** and each reflector **156** is parallel the longitudinal axis of the moderator **132**. Therefore, proximal the perimeter, the fuel elements are arranged with a small pitch offset between the axial center of each fuel element (e.g., e.g., 1.0-3.0 inches) and, distal the perimeter, the fuel elements are sparsely clustered (e.g., exhibit a high offset distance of 4-6 inches) to balance a neutron availability and thermal power output of the fuel elements throughout the reactor assembly.

1.7 Pump

The engine system **100** includes a pump **110** configured to pump propellant throughout the engine system **100**. In one implementation, the pump **110** includes a cold side **112** and a hot side **114**. The cold side **112** and the hot side **114** each include a separate turbine connected by a single axle. As hot propellant reaches the hot side **114** of the pump, the hot

propellant loses thermal energy to the hot side turbine, and that thermal energy rotates the hot side turbine, thereby rotating the axle between the hot and cold side. turbines and operating the cold side **112**. Therefore, the pump **110** pumps propellant throughout the engine system **100** at a propellant flow rate that is proportional to the amount of thermal energy provided by the propellant to the hot side **114** of the pump. In this implementation, the pump **110** can also be operated via pressurized fluid exerting force against the turbines. For example, during a startup period of the engine system **100**, an orifice of the propellant reservoir **178** is opened to release pressurized propellant to the cold side **112** of the pump, therefore operating the cold side **112** and hot side **114** of the pump **110**.

1.8 Reflectors

The engine system **100** includes an array of reflectors **156** arranged about a perimeter of the reactor assembly **120** and configured to reflect neutrons into and out of the reactor assembly **120**. The reflector **156** includes: a neutron-reflecting material; and a reflector coolant channel **170** arranged within the reflector **156** and configured to exchange thermal energy with propellant pumped through the reflector **156** coolant channel. The reflector **156** is configured to: at a first time, operate in a closed position to reflect neutrons within the reactor assembly **120** to increase an energy flux of the engine system **100**; and, at a second time, operate in an open position configured to release neutrons out of the set of fuel elements to decrease the energy flux of the engine system **100**.

In one implementation, the reflector **156** defines a thin rectangular prism geometry (e.g., a panel) including a longitudinal axis arranged parallel to the longitudinal axis of the moderator **132** and the reactor assembly **120**. The reflector **156** geometry can include a taper that decreases the thickness of the reflector **156** as the reflector **156** extends away from the reactor assembly **120** in the open position.

The reflector **156** couples to a housing of the engine system **100** or directly to the moderator **132** via an actuatable hinge that enables the reflector **156** to rotate about a hinge point defined by the hinge to occupy an open or closed position. In one implementation, the engine system **100** can include a single reflector **156** configured to operate in an open position to allow neutrons to leak out of the set of fuel elements **122** into the external environment to decrease a rate of the fission reaction of the fuel elements. In the cylindrical moderator **132** example, the engine system **100** includes an array of reflectors **156** configured to completely surround the moderator **132** in the closed position. The array of reflectors **156** can overlap to effectively seal neutrons in the reactor assembly **120** in the open position.

The reflector **156** includes a neutron-reflector **156** material such as beryllium. In one implementation, shown in FIG. 4B, the reflector **156** includes a cast solid beryllium panel or fin with a reflector coolant channel **170** drilled through the panel (e.g., extending within the panel). The reflector **156** can also include a beryllium ceramic material that is: 3D printed to define an interior void defining the reflector **156** coolant channel; or sintered to a target density such that the sintered particles define a tortuous coolant channel.

For example, the reflector can include a metallic beryllium structure defining: a pivot axis; a length facing the set of fuel elements in the closed configuration; a width facing the set of fuel elements in the open configuration, the width less than the length; a fluid inlet coaxial with the pivot axis and fluidly coupled to an outlet of the nozzle coolant channel; a fluid outlet coaxial with the pivot axis and fluidly

coupled to an inlet of the set of moderator coolant channels; and an interior volume a) fluidly coupled to the fluid inlet and the fluid outlet; b) defining the reflector coolant channel; and c) defining a series of internal vanes **188** intersecting the reflector coolant channel. The reflector can additionally include a reflector actuator **186**: coupled to the reflector; and configured to pivot the reflector about the pivot axis between the open configuration and the closed configuration.

In one implementation, the engine system can include a first reflector and a second reflector including an electric motor (e.g., the reflector actuator **186**) coupled to the first reflector and to the second reflector via a timing belt and is configured to concurrently: pivot the first reflector about the pivot axis between the open configuration and the closed configuration; and pivot the second reflector about the second pivot axis between the open configuration and the closed configuration.

In another implementation, shown in FIG. 4A, the reflector **156** can include: a rigid external shell **158**; and a set of beryllium particles **160** contained within the external shell **158**. In this implementation, the set of beryllium particles **160** define a set of interstitial volumes that define the reflector **156** coolant channel. Further, the rigid external shell **158** is configured to define a threshold melting temperature (such as during an atmospheric re-entry event) at which the rigid shell **158** can decouple from the engine system **100**, thereby opening the set of fuel elements **122** to the external environment to leak neutrons out of the set of fuel elements **122** in order to cool the set of fuel elements **122**. In both this implementation and the solid beryllium implementation, the reflector includes a coolant inlet **166** and a coolant outlet **168**. The coolant inlets and coolant outlets can define swivel connections to allow a fluid circuit to connect to the inlet and outlet while the reflector occupies different positions about the axle.

For example, the reflector can include a rigid shell defining: a pivot axis; a fluid inlet coaxial with the pivot axis and fluidly coupled to the nozzle coolant channel; a fluid outlet coaxial with the pivot axis and fluidly coupled to the set of moderator coolant channels; and an interior volume: a) fluidly coupled to the fluid inlet and the fluid outlet, b) defining the reflector coolant channel, c) defining a length facing the set of fuel elements in the closed configuration, and d) defining a width facing the set of fuel elements in the open configuration, the width less than the length. In this example, the reflector includes a matrix of beryllium particles arranged in the interior volume and configured to reflect neutrons; and is configured to pivot about the pivot axis between the open configuration and the closed configuration.

The reflector **156** couples to the engine system **100** via an axle **162** arranged on an edge of the reflector **156** proximal the reactor assembly **120**. The reflector **156** actuates via a motor that actuates a drive belt **164** mechanically coupled to the axle **162**. The axle **162** can include a set of teeth configured to engage with a set of teeth of the drive belt **164**. Therefore, as the motor actuates the drive belt **164**, the axle **162** rotates, thereby rotating the reflector.

In one implementation, a motor includes a set of drive belts **164** including a drive belt **164** coupled to each reflector **156** of an array of reflectors **156**. For example, the motor can actuate 12 drive belts **164** coupled to 12 reflectors **156**. In this implementation, the motor actuates the array of reflectors **156** together.

In another implementation, the engine system **100** includes a set of motors, wherein each motor is configured to actuate an array of reflectors **156**. For example, the engine

system **100** can include a set of three motors, each motor configured to actuate four drive belts **164** coupled to four reflectors **156**. Therefore, the set of twelve reflectors **156** are actuatable as three separate groups.

In another implementation, the engine system **100** includes a set of motors, wherein each motor is configured to actuate a single drive belt **164** coupled to a single reflector. For example, the engine system **100** can include a set of 12 motors, each motor configured to operate a single reflector **156** of the set of 12 reflectors **156**.

1.9 Thrust Chamber & Thrust Nozzle

The engine system **100** can further include a thrust chamber **176** and a thrust nozzle **152** configured to: receive heated propellant output by the set of fuel elements; and release the propellant out of the engine system **100** through a nozzle to produce thrust. The thrust nozzle **152** and thrust chamber **176** are arranged downstream of the reactor assembly **120**. The thrust chamber **176** receives heated propellant from each fuel element **122**. Propellant flows from different fuel elements and can mix in the thrust chamber **176** such as to equalize to a single temperature. The thrust nozzle **152** is configured to release propellant from the reactor assembly **120** to produce thrust.

The thrust nozzle **152** and thrust chamber **176** can increase in temperature due to the thermal energy of the propellant within these components. Therefore, the thrust nozzle **152** includes a nozzle coolant channel **154** arranged within a wall of the thrust nozzle **152** and configured to exchange thermal energy with propellant pumped through the nozzle coolant channel **154**. The thrust chamber **176** can include a chamber coolant channel within a wall of the chamber configured to cool the chamber with a flow of propellant.

1.10 Fluid Circuits

The components of the engine system **100** described above are joined via a fluid circuit **190** configured to transport propellant between components. The fluid circuit **190** can include a set of pipes and a set of valves connecting the aforementioned components. For example, the fluid circuit **190** directs propellant: from the propellant reservoir to the cold side **112** of the pump; to the nozzle coolant channel **154** and reflector **156** coolant channel; to the moderator coolant channels **134**; to the hot side **114** of the pump; through the reactor assembly **120**; and into the thrust chamber **176** to be outlet by the thrust nozzle. The fluid circuit **190** can include a set of valves configured to moderate flows of propellant to each component. The set of valves can additionally open and close bypass pathways within the fluid circuit **190** to direct propellant directly to a component (e.g., by bypassing another component). For example, a valve can open responsive to a high pressure within the fluid circuit **190** to bypass the nozzle coolant channel **154** and direct a flow of propellant to the reflector **156** coolant channel. The set of valves can additionally be electronically actuated by a controller to divide a flow of propellant between two fluid circuit pathways or open/close a pathway of the fluid circuit.

In one implementation, the fluid circuit **190** of the engine system **100** can include: a coolant supply path **192**; a thrust path **194**; and a coolant return path **196**. The coolant supply path **192** includes a fluid circuit: from the propellant reservoir **178**; to the cold side **112** of the pump; to the nozzle coolant channel **154**; to the reflector **156** coolant channel;

11

and into the set of moderator coolant channels **134**. The coolant supply path **192** can include: a first coolant bypass valve configured to a) operate in an open configuration to pass propellant through the nozzle coolant channel **154**, and b) operate in a closed configuration to block propellant from entering the nozzle coolant channel **154** and pass the propellant directly into the reflector coolant channel **170** and a second coolant bypass valve configured to a) operate in an open configuration to pass propellant through the reflector **156** coolant channel, and b) operate in a closed configuration to block propellant from entering the reflector coolant channel **170** and pass the propellant directly into the set of moderator coolant channels **134**.

The thrust path **194** can include the fluid circuit: extending from the moderator coolant channels **134**; to the hot side **114** of the pump; through the set of fuel elements; and out of the thrust nozzle. In one implementation, the thrust path can include a coolant return path **196** between an outlet **180** of the moderator coolant channels **134** and the hot side **114** of the pump. The thrust path **194** can additionally include: a pump bypass valve **172** configured to a) operate in an open position to pass propellant from the outlet of the set of moderator coolant channels **134** to the hot side **114** of the pump, and b) operate in a closed position to block the hot side **114** of the pump **110** thereby passing the propellant from the outlet of the set of moderator coolant channels **134** to the set of fuel elements **122**.

1.11 Sensors

The engine system **100** can additionally include a set of sensors configured to output signals indicating conditions of the engine system **100**. The set of sensors can include accelerometers, temperature sensors, mechanical load sensors, pressure sensors, neutron sensors, flow sensors, and position sensors. For example, engine system **100** can include a temperature sensor arranged within each fuel element **122** and configured to output a signal representing a temperature within each fuel element **122**. The pressure sensors can arrange throughout the fluid circuit to output signals corresponding to a pressure within the fluid circuit to the controller. The position sensors can detect a position of a valve and/or a reflector **156** of the engine system **100**. The set of sensors can include both electronic and mechanical sensors.

1.12 Controller

The engine system **100** includes a controller configured to: access target parameters (e.g., target temperatures, target thrusts, target neutron populations); receive signals from sensors; actuate a reflector via a reflector actuator **186**; and actuate valves within the engine system **100**.

In one implementation, the controller is arranged within the payload to reduce radiation damage to the electronics within the controller. In another arrangement, the controller is located within or proximal the engine system **100** and includes a shield.

The controller is configured to: access target parameters of the system (e.g., temperatures, pressures, angles, neutron populations, and thrusts); and implement closed-loop controls to shift the current parameters of the system toward the target parameters. For example, the controller can: access a target fuel element **122** temperature associated with a current target thrust of the engine system **100**; and read an output signal from a temperature sensor near the fuel elements. In response to the output signal from the temperature

12

sensor of the fuel element **122** indicating a temperature higher than the target fuel element temperature, the controller actuates a pump bypass valve **172** to direct additional flow of propellant from the moderator **132** coolant outlet to the hot side **114** of the pump **110** to increase a propellant flow rate of propellant through the fuel elements, thereby lowering the temperature of the fuel elements. The controller can replicate these steps to access other target parameters and adjust temperatures, pressures, and fission rates, throughout the engine system **100**.

1.12.1 Reflector Position

In one implementation, the reflectors **156** are coupled to a housing of the engine system via reflector axles **162** and actuated by a reflector actuator **186**. The engine system can include a set of reflector position sensors (e.g., a splined encoder, a magnetic encoder, a depth sensor) configured to output a signal to the controller corresponding to a current position and/or angle of each reflector. The controller can therefore access the reflector position sensor signal to detect the position of the reflectors and calculate an angle to trigger the reflector actuator **186** to which it can actuate the reflectors **156**.

In one example wherein the moderator defines a cylindrical geometry, the set of fuel elements define a circular array within the moderator, and the reflector couples to a hinge point of a housing of the moderator proximal the perimeter of the cylindrical geometry. In response to a first target thrust, the reflector occupies (e.g., the controller triggers the reflector actuator to actuate the reflector to) a first shallow angle relative to a tangent of the perimeter of the set of fuel elements such that a widest cross-section of the neutron-reflective material of the reflector faces the reactor assembly **120** and reflects a greater number of neutrons back into the set of fuel elements **122** to increase the rate of fission within the set fuel elements to a first target rate of fission proportional to the first target thrust. In response to a second target thrust less than the first target thrust, the reflector occupies a second wide angle, greater than the first shallow angle, relative to the tangent of the perimeter of the reactor assembly such that a narrowest cross-section of the neutron-reflective material of the reflector faces the set of fuel elements, releases more neutrons to an external environment to reduce the rate of fission within the reactor assembly to a second target rate of fission proportional to the second target thrust. In one implementation, the controller: calculates a target angle of the reflectors **156** based on a current thrust output of the system, a target thrust output of the system, a current temperature of the fuel elements, and/or a target a temperature of the fuel elements.

In one implementation, the controller detects an instantaneous thrust from outputs of accelerometers and mechanical load sensors. The controller can then: calculate an integrated total impulse and detect, based on the integrated total impulse if a target impulse condition is met (e.g., a target impulse has been produced by the engine system).

In response to the current thrust greater than the target thrust, the controller: triggers the reflector actuator to open the reflectors **156** by angular distance proportional to the difference between the current thrust and the target thrust, thereby releasing neutrons from the system, reducing a rate of fission within the fuel elements, reducing temperatures within the fuel elements, and reducing the current thrust output from the system.

In response to the current thrust less than the target thrust, the controller: triggers the reflector actuator to close the reflectors **156** by angular distance proportional to difference

13

between the current thrust and the target thrust, thereby increasing the availability of neutrons within the system, increasing a rate of fission within the fuel elements, increasing temperatures within the fuel elements, and increasing the current thrust output from the system.

1.12.2 Thrust-Coolant Valve Position

The engine system includes a thrust-coolant valve **174** configured to actuate direct propellant flow from the cold side of the pump to the moderator coolant channels or to the nozzle coolant channel. The thrust-coolant valve can regulate both the rate of propellant flow from the cold side of the pump and the proportion of propellant flow directed to the moderator coolant channel and nozzle coolant channel. The thrust-coolant valve is configured to actuate to a set of positions to divide a flow of propellant from the cold side of the pump between the moderator coolant channel and the nozzle coolant channel.

For example, the controller triggers the thrust-coolant valve **174** to actuate to a first position in which the thrust-coolant valve directs 100% of the propellant from the cold side of the pump to the moderator coolant channel to cool the moderator. The controller can trigger the thrust-coolant valve to actuate to a second position in which the thrust-coolant valve directs 100% of the propellant from the cold side of the pump to the nozzle coolant channels to cool the nozzle and reflectors.

The thrust-coolant valve is additionally configured to actuate between the first and second positions. For example, in response to the controller detecting a temperature of the moderator over the moderator operating temperature range, the controller triggers the thrust-coolant valve to actuate to increase a proportion of flow from the propellant to the moderator coolant channel and decrease the proportion of propellant flowing to the nozzle coolant channels. In this example, if the thrust-coolant valve initially occupied a position dividing the propellant such that 50% of the propellant flowed to the moderator coolant channel and 50% of the propellant flowed to the nozzle coolant channel, the controller can actuate the thrust-coolant valve to divide the propellant to direct 75% of the flow to the moderator coolant channel and 25% of the flow to the nozzle coolant channel.

In response to the controller detecting a temperature of the nozzle over the nozzle operating temperature range, the controller can trigger the thrust-coolant valve to actuate to increase the amount of propellant flowing to the nozzle coolant channel and decrease the amount to the moderator coolant channel.

Increasing the flow of propellant to the nozzle coolant channels decreases the temperature of the propellant that enters the moderator coolant channels, thereby decreasing the temperature of propellant that exits the moderator coolant outlet **180** and flows to the fuel elements.

1.12.3 Pump Bypass Valve Position

The engine system includes a pump bypass valve **172** configured to actuate to direct propellant from the moderator coolant outlet **180** to the hot side of the pump or to the set of fuel elements. Pump bypass valve **172** is configured to actuate to a set of positions to divide the flow of propellant from the moderator coolant outlet between the hot side of the pump and the set of fuel elements.

For example, the controller triggers the pump bypass valve to actuate to a first position in which the pump bypass valve directs 100% of the propellant to the hot side of the pump. In this example, the propellant loses thermal energy to the hot side of the pump, causing the pump to increase a rotation rate at the cold and hot sides. Therefore, the flow rate within the engine system increases. Therefore, in

14

response to the controller detecting a current thrust below a target thrust, the controller can actuate the pump bypass valve to increase a proportion of propellant directed to the hot side of the pump to increase the flow rate through the set of fuel elements thereby increasing the thrust. The controller is additionally configured to actuate the pump bypass valve to decrease the proportion of propellant flowing to the hot side of the pump and increase the proportion of the propellant flowing directly to the set of fuel elements to increase a temperature of the fuel elements.

In another example, increasing the flow of propellant to the hot side of the pump increases the flow rate throughout the engine system.

The controller can trigger the pump bypass valve to actuate to balance a mass flow rate and a temperature of the engine system. For example, the controller can trigger the pump bypass valve to actuate to a position in which 25% of the flow from the moderator coolant outlet is directed to the hot side of the pump to maintain a rate of flow through the system and 75% of the flow of propellant is directed to the set of fuel elements to cool the fuel elements and produce thrust.

1.12.4 Circulation Valve

The engine system can include a circulation valve **182** configured to direct propellant to circulate through a circulation loop (FIG. 2F). During a circulation state of the engine the controller actuates the circulation valve **182** to a circulation position in which 100% of the propellant is directed into the circulation loop. In the circulation state, the propellant flows from the moderator coolant outlet **180** to the pump bypass valve that is actuated to direct 100% of the propellant to the set of fuel elements (e.g., in a full pump bypass position). The propellant travels through the circulation valve in the circulation position and into the circulation loop. The circulation loop directs the propellant to the nozzle coolant channel, to the reflector coolant channel, and back into the moderator coolant channel. The coolant loop can include an active pump (e.g., a hydraulic or electric pump) configured to pump cool propellant throughout the coolant loop during the circulation state.

1.12.5 Thrust Profile

In one implementation, the controller is configured to calculate a thrust profile for the nuclear rocket engine to produce a total; impulse. The thrust profile can include: the first time period defining a hot-thrust state characterized by locating the reflector at the closed position to increase the fission rate within the set of fuel elements and increase a total impulse output of the nuclear rocket engine to the first thrust; and the second time period defining cooldown state characterized by locating the reflector at the open position to decrease the fission rate within the set of fuel element, cool the set of fuel elements to a nominal temperature, and estimate the decreasing thrust output of the nuclear rocket engine to the second thrust less than the first thrust. In this implementation, the first time period defining the hot-thrust state period includes generating a first proportion of the target total impulse, and the second time period defining the cooldown-thrust state period includes generating a second proportion of the target total impulse, a sum of the first proportion and the second proportion approximating the target total impulse.

For example, during controller can calculate that 90% of a target total impulse is produced in the first stage while the remaining 10% is produced with the reflectors **156** open during the second stage. Therefore, the controller can

15

receive a target thrust or impulse signal and actuate reflector **156** positions to achieve a total impulse approximating the target thrust or total impulse.

2 METHOD

As shown in FIGS. 2A and 2B, a method **S100** for controlling a nuclear rocket engine system **100** includes, during first time period at the nuclear rocket engine by a pump, pumping a propellant: from a propellant reservoir **178** at a supply temperature in Block Silo; through a cold side **112** of the pump **110** in Block **S112**; to a set of moderator coolant channels **134**, extending axially within a moderator **132**, to cool the moderator **132** in Block **S114**; to a hot side **114** of the pump **110** to operate the cold side **112** of the pump **110** in Block **S116**; through a set of fuel elements to heat the propellant to a first outlet temperature, the set of fuel elements extending axially within the moderator **132** and isolated from the set of moderator coolant channels **134** in Block **S118**; and out of a thrust nozzle **152** to produce a first thrust in Block **S120**. The method **S100** additionally includes, during the first time period locating the reflector, arranged about the set of fuel elements and the moderator **132**, at a first closed position to reflect neutrons toward the reactor assembly **120** to maintain a fission rate of the set of fuel elements above a threshold fission rate in Block **S122**.

Additionally, during a second time period, the method **S100** includes locating the reflector **156** at a second open position to release additional neutrons from the nuclear rocket engine and to reduce the fission rate of the set of fuel elements to below the threshold fission rate in Block **S124**. Further, during the second time period the pump, pumps the propellant: from the propellant reservoir **178** at the supply temperature in Block **126**; through the cold side **112** of the pump **110** in Block **S128**; to the set of moderator coolant channels **134** to cool the moderator **132** in Block **S130**; to the hot side **114** of pump **110** to operate the first side of the pump **110** in Block **S132**; through the set of fuel elements to cool the set of fuel elements and to heat the propellant to a second outlet temperature less than the first outlet temperature in Block **S134**; and out of the thrust nozzle **152** to produce a second thrust less than the first thrust in Block **S136**.

In one variation of the method **S100**, during the first time, the pump **110** pumps propellant: from a propellant reservoir **178** at a supply temperature; through a cold side **112** of the pump; to a set of moderator coolant channels **134**, extending axially within a moderator **132**, to cool the moderator **132**; to a hot side **114** of the pump **110** to operate the cold side **112** of the pump; through a cold shell **124** perforation of a fuel element **122** in a radial direction; through a fuel bed **128** of the fuel element **122** comprising a set of nuclear fuel particles defining a set of interstitial volumes through which the propellant flows to heat the propellant to a first outlet temperature; through a hot shell **126** perforation of the fuel element **122** in the radial direction; out of the fuel element **122** in an axial direction parallel to a longitudinal axis of the first fuel element **122**; and out of a thrust nozzle **152** to produce a first thrust.

In this variation of the method **S100**, during the second period the pump **110** pumps propellant: from the propellant reservoir **178** at the supply temperature; through the cold side **112** of the pump; to the set of moderator coolant channels **134** to cool the moderator **132**; to the hot side **114** of pump **110** to operate the first side of the pump; through the cold shell **124** perforation of the fuel element **122** in the radial direction; through the fuel bed **128** of the fuel element

16

122 to heat the propellant to a second outlet temperature less than the first outlet temperature; through the hot shell **126** perforation of the fuel element **122** in the radial direction; out of the fuel element **122**; and out of the thrust nozzle **152** to produce a second thrust less than the first thrust.

In another variation of the method **S100**, the method includes, during a first time period at a nuclear rocket engine locating the reflector, arranged about a set of fuel elements at a first angle to: reflect a first proportion of neutrons toward the set of fuel elements to maintain a first fission rate of the set of fuel elements above a threshold fission rate; and reflect a second proportion, less than the first proportion, of neutrons out of the nuclear rocket engine. Additionally at the first time period, the pump **110** pumps propellant: to a set of moderator coolant channels **134**, extending axially within a moderator **132**, to cool the moderator **132**; through the set of fuel elements to heat the propellant to a first outlet temperature proportional to the first fission rate, the set of fuel elements extending axially within the moderator **132** and isolated from the set of moderator coolant channels **134**; and out of a thrust nozzle **152** to produce a first thrust.

In this variation of the method **S100**, the method **S100** includes during a second time period locating the reflector **156** at a second angle to: reflect a third proportion of neutrons toward the set of fuel elements to reduce a second fission rate of the set of fuel elements to below the threshold fission rate; and reflect a fourth proportion, greater than the third proportion, of neutrons out of the nuclear rocket engine. Additionally, the pump **110** pumps propellant: to the set of moderator coolant channels **134** to cool the moderator **132**; through the set of fuel elements to cool the set of fuel elements and to heat the propellant to a second outlet temperature less than the first outlet temperature and proportional to the second fission rate; and out of the thrust nozzle **152** to produce a second thrust less than the first thrust.

2.1 Applications

Generally, the method **S100**, includes: a hot thrust-producing state; and a cooldown state. During the hot thrust-producing state: the reflectors **156** are actuated to a closed position to reflect neutrons toward the set of fuel elements; and the pump **110** pumps propellant through the moderator coolant channels **134** to cool the moderator **132**, through a set of fuel elements to heat the propellant to a first outlet temperature, and out of a thrust nozzle **152** to produce a first thrust. During the cooldown state: the reflectors **156** are actuated to an open position to allow neutrons to exit the set of engine system **100** to the external environment; and the pump **110** pumps propellant through the moderator coolant channel **134** to cool the moderator **132**, through the set of fuel elements to heat the propellant to a second outlet temperature less than the first outlet temperature, and out of the thrust nozzle **152** to produce a second thrust less than the first thrust. The method **S100** can transition the engine system **100** to additional states including: a virgin startup state; a bleed-off state; and a re-start state.

The method **S100** enables the nuclear rocket engine system **100** to transition states to modulate: the fission rate within the set of fuel elements; and the amount of thrust produced. Within these engine system **100** states, the engine system **100** divides flow of propellant throughout the engine system **100** to direct high flows to components that are over an operating temperature range. In this way, the method

preserves the longevity of the nuclear rocket engine system 100 by maintaining a target operating temperature within the engine system 100.

2.2 Engine Cycle

Generally, the engine cycle of the engine system 100 includes pumping propellant through fluid circuits of the engine system 100 and altering the positions of reflectors 156 to: produce thrust; and maintain the components of the engine system 100 within an operating temperature range. In one implementation, the engine system 100 produces thrust and maintains the operating temperature by actuating: a thrust-coolant valve 174; a pump bypass valve 172; and the array of reflectors 156.

To produce thrust the pump 110 pumps propellant: through the set of fuel elements arranged within the moderator 132; and out of the thrust nozzle. Pumping propellant through the set of fuel elements further includes pumping, by the pump, the propellant: through a cold shell 124 perforation of a first fuel element 122 of the set of fuel elements in a radial direction; through a fuel bed 128 of the first fuel element 122 comprising a set of low-enriched uranium nuclear fuel particles defining a set of interstitial volumes through which the propellant flows; through a hot shell 126 perforation of the first fuel element 122 in the radial direction; and out of the first fuel element 122 in an axial direction parallel to a longitudinal axis of the first fuel element 122. The engine system 100 produced thrust by heating a volume of propellant by directing the propellant through the fuel bed 128 of a fuel element 122 to absorb thermal energy from the fission reaction of the nuclear fuel particles. In one implementation, the engine system 100 can increase thrust by directing propellant from the outlet 180 of the moderator coolant channels 134 to the hot side 114 of the pump 110 to rotate the pump 110 faster, thereby increasing a propellant flow rate of propellant pumped through the fuel elements.

To maintain the components of the engine system 100 within the operating temperature range, the pump 110 pumps cool propellant (e.g., at the supply temperature, as low as 14K) through the components of the engine system 100. For example, to cool the moderator 132 to a temperature below the maximum operating temperature of the moderator 132, the pump 110 pumps propellant: from the propellant reservoir 178; through the cold side 112 of the pump; and to the moderator coolant channels 134. The pump 110 additionally pumps propellant: from the propellant reservoir 178; to the nozzle coolant channel 154 to cool the thrust nozzle; and to the reflector coolant channel 170 to cool the reflector. Additionally, the fuel bed 128 of each fuel element 122 is cooled by the pump 110 pumping propellant through the fuel element 122, thereby preventing meltdown of the nuclear fuel particles.

The controller of engine system 100 can actuate a set of valves to moderate a proportion of flow of propellant pumped to each component. For example, the engine system 100 can include a thrust-coolant valve 174 arranged between the cold side 112 of the pump 110 and the coolant supply path and propellant supply path configured to: actuate to a target position to pass a first target proportion of propellant into the coolant supply path and to pass a second target proportion of propellant into the propellant supply path, wherein the first target proportion of propellant is larger than the second target proportion of propellant to increase a mass propellant flow rate of propellant, and the first target pro-

portion of propellant is smaller than the second target proportion of propellant to decrease the mass propellant flow rate of propellant.

The controller can therefore dynamically adjust: a first proportion of propellant from the cold side 112 of the pump 110 to the coolant pathway (e.g. through the nozzle coolant channel 154 and reflector 156 coolant channel); and a second proportion of propellant from the cold side 112 of the pump 110 to the thrust pathway (e.g., to the moderator coolant channels 134 and into the set of fuel element 122).

In one implementation, to decrease a temperature of the nozzle and reflector, the controller actuates the thrust-coolant valve 174 to a position such that a first proportion of propellant directed to the coolant pathway is higher than the second proportion to the thrust pathway.

2.3 Engine State: Conditioning

In one implementation, before the engine system 100 can produce thrust, the engine system conditions (e.g., heats, circulates) the propellant within the fluid circuits to a target conditioning temperature range, shown in FIGS. 2A and 6B. In the conditioning state, the pump bypass valve 172 and circulation valve 182 occupies a closed position configured to direct propellant to the fuel elements 122 (e.g., bypassing the hot side 114 of the pump 110). The thrust-coolant valve occupies an open position to direct propellant to the nozzle coolant channel and reflector coolant channel. The engine system directs propellant: from the propellant reservoir; to the cold side 112 of the pump; through the thrust-coolant valve 174; through the nozzle coolant channel and reflector coolant channel (of the open reflectors 156); through the moderator coolant channel; and through the set of fuel elements. In this state, the propellant circulates slowly through the fluid circuit to transfer thermal energy between components to reach an equilibrium state (e.g., wherein the components each occupy a target conditioning temperature range). In one implementation, the equilibrium state can include each component occupying approximately equal temperatures. Due to the low temperature and flow rate of the propellant expelled by the thrust nozzle during the conditioning state, minimal thrust (e.g., <100N) is produced.

The controller can read sensors throughout the engine system to detect if the equilibrium conditions are met. For example, the controller can read the moderator and fuel element temperatures from temperature sensors and read the flow rate throughout the fluid circuit from a flow sensor. Based on the outputs of these sensors, the controller detects the equilibrium condition and triggers the reflector actuators to begin incrementally closing the reflectors to enter the bootstrapping state (shown in FIG. 6C). If the outputs of the sensors do not indicate the equilibrium condition the controller can wait (e.g., set a timer and upon expiration of the timer) read the sensor outputs at a later time.

2.4 Engine State: Bootstrapping

During the bootstrapping state, shown in FIGS. 2B and 6C the controller triggers the reflector actuators to actuate the reflectors to close incrementally.

If the reactor assembly includes a virgin core (e.g., wherein the nuclear fuel particles are not exhibiting active fission), the controller can trigger a neutron generator to direct a stream of neutrons into the reactor assembly 120 to collide with the nuclear fuel particles within a fuel element 122 and begin the fission reaction, thereby increasing the temperature within the fuel elements 122 and reactor assem-

bly 120. During virgin startup state, there are no residual decay products, and neutron flux is effectively zero within the reactor assembly 120.

During the bootstrapping state, the controller triggers the thrust-coolant valve 174 to actuate to a position in which the thrust-coolant valve 174 directs: approximately 50% of propellant from the cold side 112 of the pump 110 to the moderator coolant channel; and approximately 50% of propellant from the cold side 112 of the pump 110 to the nozzle coolant channel and reflector coolant channel. The controller triggers the pump bypass valve 172 to actuate to direct 100% of the propellant from the outlet of the moderator coolant channels to the hot side 114 of the pump 110 to operate the pump.

In one implementation, the controller can calculate a target reflector 156 angle based on an incremental (less than full) target thrust and a current fuel element 122 temperature. The controller actuates the reflector 156 to a closed position to increase the fission rate within the set of fuel elements and begin to produce thrust. The controller monitors neutron flux (e.g., by reading the output of a neutron sensor or dosimeter) and fuel element temperatures (e.g., by reading the output of a temperature sensor near the fuel element) to detect a critical reaction state. In response to reaching the critical reaction state (e.g., defining a stable fission reaction and/or a temperature within the fuel element high enough to produce a target thrust), the controller can: trigger the thrust-coolant valve to increase a proportion of propellant flowing to the moderator coolant channel; and read a mass flow, moderator temperature, fuel element temperature, and instantaneous thrust produced to detect a bootstrap equilibrium state. During the bootstrapping state, the engine system produced an instantaneous thrust higher than the during the conditioning state, but below the hot thrust state output. Once a set of target bootstrap parameters are met (e.g., a threshold fuel element temperature, threshold flow rate etc.) the engine system can enter the hot thrust state.

2.5 Engine State: Hot Thrust

After the startup time period that engine system 100 can occupy a hot thrust state, shown in FIGS. 2C and 6D, in which the fission reaction occupies a critical state and the engine system 100 produces thrust. During the hot thrust state the reflector 156 or array of reflectors 156 occupy a closed position to reflect neutrons within the reactor assembly 120.

During the hot thrust state, the pump, pumps propellant: from a propellant reservoir 178 at a supply temperature; through a cold side 112 of the pump; to a set of moderator coolant channels 134, extending axially within a moderator 132, to cool the moderator 132; to a hot side 114 of the pump 110 to operate the cold side 112 of the pump; through a set of fuel elements to heat the propellant to a first outlet temperature, the set of fuel elements extending axially within the moderator 132 and isolated from the set of moderator coolant channels 134; and out of a thrust nozzle 152 to produce a first thrust; and locating the reflector, arranged about the set of fuel elements and the moderator 132, at a first closed position to reflect neutrons toward the set of fuel elements to maintain a fission rate of the set of fuel elements above a threshold fission rate.

During the hot thrust stage, the controller actuates the thrust-coolant valve 174 to maintain the nozzle, reflectors 156, moderator 132, and fuel elements within an operating temperature range such as by increasing or decreasing a

proportion of propellant directed to the moderator coolant channels or nozzle coolant channel as shown in FIG. 6D.

In one implementation, the thrust-coolant valve 174 actuates to direct 50-90% of the flow of propellant from the cold side of the pump to the moderator coolant channel and 10-50% of the flow to the nozzle coolant channel. Additionally, the pump bypass valve 172 actuates to direct 90-100% of the flow of propellant from the outlet 180 to the hot side of the pump and 0-10% of the flow of propellant to the set of fuel elements 122.

In one implementation, the hot thrust stage includes further comprising, pumping a first proportion of the propellant: from the cold side of the pump to a nozzle coolant channel within the thrust nozzle to cool the thrust nozzle; and from nozzle coolant channel to a reflector coolant channel within the reflector to cool the reflector. In this implementation, pumping the propellant to the set of moderator coolant channels to cool the moderator during the first time period includes by the pump: pumping the first proportion of the propellant from the reflector coolant channel to the set of moderator coolant channels; and pumping a second proportion of the propellant, greater than the first proportion, from the pump directly to the set of moderator coolant channels to cool the moderator.

In one implementation, the first proportion is inversely proportional to a moderator temperature of the moderator; and the second proportion is proportional to the moderator temperature of the moderator. Pumping the propellant to the set of moderator coolant channels to cool the moderator during the first time period includes: at an inlet to the set of moderator coolant channels, mixing the first proportion of propellant at a first temperature with the second proportion of propellant at a second temperature less than the first temperature to form a propellant mixture at a third temperature less than a present temperature threshold of the moderator. In another implementation, the first proportion is further proportional to a nozzle temperature of the nozzle and proportional to a reflector temperature of the reflector.

In one implementation, the controller calculates a total engine impulse. The engine system produces: a first proportion of the total engine impulse during the hot thrust state; and a second proportion of the total engine impulse during the cooldown state. The controller can calculate the first and second proportions of total engine impulse during these stages to confirm a target total impulse is produced throughout the engine cycle.

2.6 Engine State: Cooldown

At a second time the engine system 100 occupies a cooldown state, shown in FIGS. 2D and 6E, to reduce a temperature of the engine system 100 and decrease an output of thrust. During the cooldown state, the reflector 156 occupies an open position to leak neutrons out of the reactor assembly 120 and into the external environment (e.g., into space).

For example, the method S100 includes: locating the reflector 156 at a second open position to release neutrons from the nuclear rocket engine and to reduce the fission rate of the set of fuel elements to below the threshold fission rate; and by the pump, pumping the propellant a) from the propellant reservoir 178 at the supply temperature; b) through the cold side 112 of the pump; c) to the set of moderator coolant channels 134 to cool the moderator 132; d) to the hot side 114 of pump 110 to operate the first side of the pump; e) through the set of fuel elements to cool the set of fuel elements and to heat the propellant to a second

21

outlet temperature less than the first outlet temperature, and f) out of the thrust nozzle **152** to produce a second thrust less than the first thrust. In one implementation, the controller can actuate the thrust-coolant valve **174** to divide the flow of propellant from the cold side **112** of the pump **110** between the moderator coolant channels and the nozzle coolant channel. For example, the thrust-coolant valve can direct 70% of the flow of propellant to the moderator coolant channels and the remaining 30% to the nozzle coolant channel or the thrust-coolant valve can split the flow **50/50** as shown in FIG. 2D.

During the cooldown state the controller can actuate the pump bypass valve **172** such that the pump **110** pumps propellant from the outlet **180** of the moderator coolant channels **134** to the hot side **114** of the pump to operate the pump with the remaining thermal energy in the fluid circuit.

In one implementation, during the cooldown state, the pump can pump a proportion of propellant: from the cold side of the pump to the nozzle coolant channel within the thrust nozzle to cool the thrust nozzle; and from nozzle coolant channel to the reflector coolant channel within the reflector to cool the reflector coolant channel. Pumping the propellant to the set of moderator coolant channels to cool the moderator during the second time period includes, by the pump: pumping the third proportion of the propellant from the reflector coolant channel to the set of moderator coolant channels, the third proportion less than the first proportion; and pumping a fourth proportion of the propellant, greater than the first proportion, from the pump directly to the set of moderator coolant channels to cool the moderator, the fourth proportion greater than the second proportion, wherein a first volume of propellant comprising the first proportion and the second proportion is less than a second volume of propellant comprising the third proportion and the fourth proportion.

2.7 Engine State: Bleed-Off

In one implementation, during or after the cooldown state, the engine system can enter a bleed-off state to purge a portion of hot propellant out of the thrust nozzle, as shown in FIGS. 2E and 6F. During this state, the controller can actuate an orifice of the propellant reservoir to introduce a portion of cold propellant from the propellant reservoir **178** into the fluid circuit to lower the average temperature of the propellant and the components within the fluid circuit.

In the bleed-off state, the engine system **100** coordinates positions of the reflectors **156**, the thrust-coolant valve **174**, and the pump bypass valve **172** to reduce the fission reaction rate within fuel beds **128** of the fuel elements **122**. However, fission reactions may persist—at low rate—within the fuel elements **122** in the bleed-off state such that the fuel elements **122** continue to release thermal energy. Therefore, the engine system **100** can intermittently release heated propellant into the fuel elements and out of the nozzle **152** to carry thermal energy (i.e., waste heat) out of the system while introducing cold propellant from the propellant reservoir **178** into the thrust pathway. More specifically, the system can: circulate propellant between the moderator and the reflectors **156** to remove heat from the fuel elements and to radiate this heat into space; and intermittently displace warm propellant in the thrust pathway with cold propellant from the propellant reservoir **178**, thereby limited thermal energy buildup within the fuel elements, maintaining substantially uniform temperatures across the moderator and the fuel elements, and limiting the fission reaction rate within the fuel elements.

22

Additionally, during the bleed-off state, the controller can actuate the pump bypass valve **172** to direct a portion of propellant from the outlet **180** of the moderator coolant channels back into the propellant reservoir **178**. For example, the engine system can include a fluid circuit connecting the outlet **180** of the moderator coolant channels to the propellant reservoir **178** and the controller actuates the pump bypass valve **172** to direct 5% of the flow of propellant to the pump bypass valve to the propellant reservoir **178**. The controller actuates the pump bypass valve to direct propellant to the propellant reservoir to re-pressurize the propellant reservoir. Therefore, the propellant reservoir maintains a pressure sufficient to continue circulating propellant throughout the fluid circuit during bleed-off, without the use of an electric pump.

During the bleed-off state, controller reads the outputs of sensors to detect: the propellant temperature, the fuel element temperature, the propellant pressure, and the propellant reservoir pressure. In response to the propellant reservoir pressure over a threshold pressure limit or the propellant or fuel element temperature over a target temperature, or a propellant pressure above a target propellant pressure the controller can bleed-off a portion of propellant from the thrust nozzle and replace the portion of hot propellant with a portion of cold propellant from the propellant reservoir.

In one implementation, in response to the reservoir pressure, propellant temperature, fuel element temperature, and propellant pressure within respective target ranges, the controller can: trigger the pump bypass valve to reduce a flow of propellant to the propellant reservoir; and trigger the circulation valve to direct flow to the nozzle coolant channel.

2.8 Engine State: Circulation

After the cooldown state, the engine system **100** can enter a circulation state, shown in FIG. 2F to circulate propellant throughout the fluid circuit. During the circulation state, the reflectors **156** remain in the open position and the engine system **100** circulates propellant at a low propellant flow rate through the engine system **100** via the second pump **184** to maintain the system at a nominal temperature range. Therefore, the system can: circulate propellant between the moderator and the reflectors **156** to collect thermal energy from the fuel elements and the moderator through convection with the circulating propellant and to release the energy into space via radiation through the reflectors **156**.

The circulation state includes: triggering a thrust-coolant valve **174** to transition to a full coolant position, the thrust-coolant valve in the full coolant position configured to direct propellant to the nozzle coolant channel; triggering a pump bypass valve to transition to a full pump bypass position, the pump bypass valve in the full pump bypass position configured to direct propellant to a circulation valve **182**; triggering the circulation valve **182** to transition to a circulation position, the circulation valve **182** in the circulation position configured to direct propellant to the thrust-coolant valve; and activating the second pump **184**. The second pump **184**, circulates the propellant: through the nozzle coolant channel; through the reflector coolant channel; through the moderator coolant channel; through the pump bypass valve in the full pump bypass position; through the circulation valve **182** in the open circulation position; through the thrust-coolant valve in the full coolant position; and back through the nozzle coolant channel. No thrust is produced in the circulation state.

2.9 Engine State: Re-Start

Following a period of dormancy or quiescence, the engine system **100** can enter a re-start state to increase the thrust

23

output by the engine system **100**. In the re-start state, the reflectors **156** are actuated in the open position. Cold propellant is directed from the reservoir **178** through the thrust-coolant valve **174** and circulated through the nozzle coolant channel **154**, reflectors **156**, and moderator **134**, subsequently redirected by the bypass valve **172** through the fuel elements **122** and thrust nozzle **152** establishing a temperature and pressure equilibrium within the engine. The controller then triggers the engine system to enter the conditioning state shown in FIG. 6B. Additional Controls

As shown in FIGS. 6A, 6B, 6C, 6D, and 6E the controller implements a series of closed loop controls to maintain the components of the system within a target operating temperature range and produce a target thrust. The engine system can transition between a set of engine states shown in FIG. 6A, including: an initial conditioning state shown in FIG. 6B, a bootstrapping state shown in FIG. 6C, a hot thrust state shown in FIG. 6D, a cooldown state shown in FIG. 6E, and a bleed-off state shown in FIG. 6F.

The controller can access a target thrust; access a temperature of a fuel element in the set of fuel elements; access a flow rate through the first element; calculate a current thrust based on the temperature of the fuel element and the flow rate through the fuel element; and access a current angle of the reflector. Responsive to the current thrust less than the target thrust the controller can trigger a reflector actuator to actuate the reflector to a second angle greater than the current angle to release more neutrons to an external environment to decrease the current thrust to the target thrust. Responsive to the current thrust greater than the target thrust, the controller can trigger the reflector actuator **186** to actuate the reflector to a third angle lower than the current angle to reflect more neutrons into the set of fuel elements to increase the current thrust to the target thrust.

To actuate the reflector from the current angle to a second angle the controller can: calculate a target energy flux proportional to the target thrust; calculate a second angle of the reflector defining a second cross-section, greater than a first cross-section at the current angle, of neutron-reflecting material facing the set of fuel elements the second cross-section of neutron-reflecting material reflecting a frequency of incident neutrons into the set of fuel element to achieve the target energy flux; and trigger the actuator to actuate the reflector from the current angle to the second angle.

The controller can additionally adjust the temperature of a component of the engine system **100** to a target temperature based on a target thrust. For example, if the target thrust defines a maximal thrust, the controller can calculate a target temperature for the fuel element **122** near a maximal fuel element **122** temperature to produce the target thrust. The controller can additionally calculate a target temperature for the moderator **132** below within the operating temperature range of the moderator **132** that allows the fuel element **122** to maintain the maximal fuel element **122** temperature. The controller can: access the target thrust value; read a current value of the temperature of the component from a signal output by a temperature sensor of that component; and actuate a valve to increase or decrease the current temperature to the target temperature of the component.

The controller can actuate the pump bypass valve **172** to adjust the propellant flow rate of propellant through the engine system **100**. For example, the pump **110** pumps propellant: from the propellant reservoir **178** occupying the supply temperature; through the cold side **112** of the pump **110** operating at a first rotation rate thereby defining a first propellant flow rate of the propellant; directly to the set of moderator coolant channels **134** at the supply temperature to

24

cool the moderator **132** and heat the propellant to a moderator **132** exhaust temperature; to the hot side **114** to release thermal energy to the hot side of the pump to increase the first rotation rate to a second rotation rate of the cold side of the pump and the hot side of the pump to output propellant at a second flow rate greater than the first flow; and through the set of fuel elements occupying an element entry temperature less than the moderator exhaust temperature and the second flow rate higher than the first flow rate to cool a set of nuclear fuel particles within each fuel element of the set of fuel elements. Therefore, by actuating the pump bypass valve **172**, the controller cools the set of nuclear fuel particles with a higher propellant flow rate of propellant.

In another example, the pump **110** pumps propellant, during a third time period: from an outlet of the set moderator coolant channels **134**; through a pump bypass valve **172** without operating the pump; directly to the set of fuel elements at a first propellant flow rate; and out of the thrust nozzle to produce a third thrust. During a fourth time period, the pump **110** pumps propellant: from the outlet of the set of moderator coolant channels **134**; through the pump bypass valve **172** without operating the pump; directly to the set of fuel elements at a second propellant flow rate lower than the first propellant flow rate; and out of the thrust nozzle to produce a fourth thrust less than the third thrust. Therefore, the propellant flow rate within the engine system **100** decreases due to the position of the pump bypass valve **172** allowing propellant to bypass the hot side **114** of the pump.

Further, responsive to a temperature of the reflector above a reflector operating temperature range, the controller can: trigger a nozzle coolant bypass valve to transition to a reflector coolant position configured to direct propellant from the cold side of the pump to the reflector coolant channel; and pump propellant a) from the propellant reservoir, b) through the cold side of the pump at a first flow rate, c) through a nozzle coolant bypass valve, and d) directly to the reflector coolant channel to cool the reflector.

3 CONCLUSION

As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the embodiments of the invention without departing from the scope of this invention as defined in the following claims.

We claim:

1. A system comprising:

a pump configured to pump a propellant;
a set of fuel elements, each fuel element of the set of fuel elements comprising:

a set of low enriched uranium fuel particles defining a set of interstitial volumes through which the pump pumps the propellant;

a moderator:

defining a set of fuel bores, each fuel bore in the set of fuel bores:

fluidly coupled to a warm propellant inflow manifold;

occupied by a fuel element; and

configured to direct warm propellant axially downward from the warm propellant inflow manifold along a length of the fuel element; and

defining a set of moderator coolant channels, each moderator coolant channel:

fluidly isolated from the set of fuel bores;

defining a set of moderator coolant inlets fluidly coupled to a thrust-coolant valve;

25

defining a set of moderator coolant outlets fluidly coupled to a pump bypass valve; and configured to pass propellant through the moderator to cool the moderator;

a reflector arranged on a perimeter of the moderator: 5 comprising:

- a neutron-reflecting material; and
- a reflector coolant channel arranged within the reflector and configured to exchange thermal energy with propellant pumped through the reflector coolant channel; 10

operable in:

- a first position, the reflector defining a first cross-section of the neutron-reflecting material facing the set of fuel elements in the first position and reflecting incident neutrons toward the set of fuel elements at a first rate; 15
- a second position, the reflector defining a second cross-section of the neutron-reflecting material facing the set of fuel elements in the second position and reflecting incident neutrons toward the set of fuel elements at a second rate, the second cross-section less than the first cross-section, the second rate less than the first rate; and 20

a thrust nozzle configured to outlet propellant to produce thrust; and 25

comprising a nozzle coolant channel arranged within a wall of the thrust nozzle and configured to exchange thermal energy with propellant pumped through the nozzle coolant channel. 30

2. The system of claim 1, wherein:

each fuel bore is configured to direct propellant axially through the moderator; and

each fuel element of the set of fuel elements defines a cylindrical geometry of concentric layers comprising: 35

- a cold shell defining a first array of perforations configured to direct the propellant radially through the cold shell;
- a fuel bed containing the set of low enriched uranium fuel particles; 40
- a hot shell defining a second array of perforations configured to direct the propellant radially through the hot shell; and
- an interior chamber configured to direct propellant axially out of the fuel element toward the thrust nozzle. 45

3. The system of claim 1, further comprising a propellant reservoir:

- arranged upstream of the pump;
- interposed between the set of fuel elements and a payload; 50 and
- configured to store a volume of the propellant comprising hydrogen, the volume of the propellant absorbing neutrons emitted axially by the set of fuel elements during a fission reaction to shield the payload from radiation. 55

4. The system of claim 1, wherein the moderator: further comprises a solid polymer characterized by a maximum operating temperature below which the moderator occupies a solid state.

5. The system of claim 1, wherein the set of fuel elements 60 comprises:

- a first cluster of fuel elements arranged in a first low-density distribution about a center of the moderator and offset by a first pitch distance between axial centers of the first cluster of fuel elements; 65
- a second cluster of fuel elements arranged in a first moderate-density distribution about the first cluster of

26

fuel elements and offset by a second pitch distance—between axial centers of the second cluster of fuel elements—less than the first pitch distance; and

a third cluster of fuel elements arranged in a first high-density distribution about the second cluster of fuel elements and offset by a third pitch distance—between axial centers of the third cluster of fuel elements—less than the second pitch distance.

6. The system of claim 5:

- a first cluster of moderator coolant channels arranged in a second high-density distribution about the center of the moderator and offset by the first pitch distance between axial centers of each moderator coolant channel;
- a second cluster of moderator coolant channels arranged in a second moderate-density distribution about the first cluster of moderator coolant channels and offset by the second pitch distance—between axial centers of each moderator coolant channel—greater than the first pitch distance; and
- a third cluster of moderator coolant channel arranged in a second low-density distribution about the second cluster of moderator coolant channels and offset by the third pitch distance—between axial centers of each moderator coolant channel—greater than the second pitch distance.

7. The system of claim 1:

wherein the reflector comprises a metallic beryllium structure defining:

- a pivot axis;
- a length facing the set of fuel elements in the closed position;
- a width facing the set of fuel elements in the open position, the width less than the length;
- a fluid inlet coaxial with the pivot axis and fluidly coupled to an outlet of the nozzle coolant channel;
- a fluid outlet coaxial with the pivot axis and fluidly coupled to an inlet of the set of moderator coolant channels; and
- an interior volume: 75 fluidly coupled to the fluid inlet and the fluid outlet; defining the reflector coolant channel; and defining a series of internal vanes intersecting the reflector coolant channel; and

further comprising a reflector actuator:

- coupled to the reflector; and
- configured to pivot the reflector about the pivot axis between the open position and the closed position.

8. The system of claim 7:

further comprising a second reflector:

wherein the second reflector comprises a second metallic beryllium structure defining:

- a second pivot axis;
- a second length facing the set of fuel elements in the closed position; and
- a second width facing the set of fuel elements in the open position, the second width less than the second length;
- a second fluid inlet coaxial with the second pivot axis and fluidly coupled to an outlet of the nozzle coolant channel;
- a second fluid outlet coaxial with the second pivot axis and fluidly coupled to an inlet of the set of moderator coolant channels; and
- a second interior volume: 85 fluidly coupled to the second fluid inlet and the second fluid outlet; defining a second reflector coolant channel; and

27

defining a second series of internal vanes intersecting the second reflector coolant channel; and further comprising the reflector actuator: comprising an electric motor coupled to the reflector and to the second reflector via a timing belt and configured to concurrently: pivot the reflector about the pivot axis between the open position and the closed position; and pivot the second reflector about the second pivot axis between the open position and the closed position.

9. The system of claim 1, wherein the reflector: further comprises a rigid shell defining: a pivot axis; a fluid inlet coaxial with the pivot axis and fluidly coupled to the nozzle coolant channel; a fluid outlet coaxial with the pivot axis and fluidly coupled to the set of moderator coolant channels; an interior volume: fluidly coupled to the fluid inlet and the fluid outlet; defining the reflector coolant channel; and defining a length facing the set of fuel elements in the closed position; and defining a width facing the set of fuel elements in the open position, the width less than the length; and comprises a matrix of beryllium particles arranged in the interior volume and configured to reflect neutrons; and is configured to pivot between the open position and the closed position about the pivot axis.

10. The system of claim 1: wherein the moderator defines a cylindrical geometry, the set of fuel elements defines a circular array within the moderator, and the reflector couples to a hinge point of a housing of the moderator proximal the perimeter of the cylindrical geometry; wherein, in the first position, the reflector occupies a first shallow angle relative to a tangent of the perimeter of the moderator such that a widest cross-section of the neutron-reflecting material of the reflector faces the set of fuel elements and reflects a first proportion of neutrons back into the set of fuel elements to increase a frequency of neutron-neutron collisions and increase a rate of fission within the set of fuel elements; and wherein, in the second position, the reflector occupies a second wide angle, greater than the first shallow angle, relative to the tangent of the perimeter of moderator such that a narrowest cross-section of the neutron-reflecting material of the reflector faces the set of fuel elements, and reflects a second proportion of neutrons less than the first proportion of neutrons back into the set of fuel elements to reduce a frequency of neutron-neutron collisions and reduce a rate of fission within the set of fuel elements.

11. The system of claim 1: wherein the moderator defines a cylindrical geometry, the set of fuel elements defines a circular array within the moderator, and the reflector couples to a hinge point of a housing of the moderator proximal the perimeter of the cylindrical geometry; wherein, in response to a first target thrust, the reflector: occupies a first shallow angle relative to a tangent of the perimeter of the moderator such that a widest cross-section of the neutron-reflecting material of the reflector faces the set of fuel elements; and reflects neutrons back into the set of fuel elements to increase a rate of fission within the set of fuel elements to a first target rate of fission proportional to the first target thrust; and

28

wherein, in response to a second target thrust less than the first target thrust, the reflector: occupies a second wide angle, greater than the first shallow angle, relative to the tangent of the perimeter of the moderator such that a narrowest cross-section of the neutron-reflecting material of the reflector faces the set of fuel elements; and releases neutrons out to an external environment elements to reduce the rate of fission within the set of fuel elements to a second target rate of fission proportional to the second target thrust.

12. The system of claim 1: wherein the pump pumps propellant through a coolant pathway comprising: a first fluid circuit from a cold side of the pump to the nozzle coolant channel to the reflector coolant channel to the set of moderator coolant channels; wherein the pump pumps propellant through a thrust pathway comprising: a second fluid circuit from the cold side of the pump to the set of fuel elements; and further comprising a thrust-coolant valve arranged between the cold side of the pump and the coolant supply path and propellant supply path configured to: actuate to a target position to: pass a first target fraction of propellant into the coolant pathway; and pass a second target fraction of propellant into the thrust pathway.

13. The system of claim 12: wherein, at the target position, the thrust-coolant valve passes the first target fraction of propellant, larger than the second target fraction of propellant, into the coolant pathway to increase a temperature of the propellant directed to a hot side of the pump to increase a mass flow rate of propellant.

14. The system of claim 12: wherein the coolant pathway further comprises: a first coolant bypass valve configured to: operate in an open position to pass propellant through the nozzle coolant channel; and operate in a closed position to block propellant from entering the nozzle coolant channel and pass the propellant directly into the reflector coolant channel; and a second coolant bypass valve configured to: operate in an open position to pass propellant through the reflector coolant channel; and operate in a closed position to block propellant from entering the reflector coolant channel and pass the propellant directly into the set of moderator coolant channels.

15. The system of claim 12, further comprising a coolant return pathway comprising: a third fluid circuit between an outlet of the set of moderator coolant channels and a hot side of the pump; and the pump bypass valve configured to: operate in an open position to pass propellant from the outlet of the set of moderator coolant channels to the hot side of the pump; and operate in a closed position to block the hot side of the pump to pass the propellant from the outlet of the set of moderator coolant channels to the set of fuel elements.

29

16. A system comprising:
 a pump configured to pump a propellant from a propellant reservoir;
 a set of fuel elements, each fuel element in the set of fuel elements defining a cylindrical geometry of concentric layers comprising:
 a cold shell defining a first array of perforations configured to direct the propellant radially through the cold shell;
 a fuel bed containing a set of low enriched uranium fuel particles;
 a hot shell defining a second array of perforations configured to direct the propellant radially through the hot shell; and
 an interior chamber configured to direct propellant axially out of the fuel element toward a thrust nozzle;
 a moderator:
 defining a set of fuel bores, each fuel bore in the set of fuel bores:
 fluidly coupled to a warm propellant inflow manifold;
 occupied by a fuel element; and
 configured to direct warm propellant axially downward from the warm propellant inflow manifold along a length of the fuel element; and
 defining a set of moderator coolant channels, each moderator coolant channel:
 fluidly isolated from the set of fuel bores;
 defining a set of moderator coolant inlets fluidly coupled to the thrust-coolant valve;
 defining a set of moderator coolant outlets fluidly coupled to the pump bypass valve; and
 configured to pass propellant through the moderator to cool the moderator;
 a reflector arranged on a perimeter of the moderator:
 comprising:
 a neutron-reflecting material; and
 a reflector coolant channel arranged within the reflector configured to exchange thermal energy with propellant pumped through the reflector coolant channel;
 operable in:
 a first position to reflect a first proportion of incident neutrons toward the set of fuel elements; and
 a second position to release a second proportion of incident neutrons to an external environment; and
 the thrust nozzle configured to outlet propellant to produce thrust:
 comprising a nozzle coolant channel arranged within a wall of the thrust nozzle configured to exchange thermal energy with propellant pumped through the nozzle coolant channel.

17. The system of claim 16:
 wherein the moderator defines a cylindrical geometry, the set of fuel elements defines a circular array within the moderator, and the reflector couples to a hinge point of a housing of the moderator proximal the perimeter of the cylindrical geometry;
 wherein, in the first position, the reflector occupies a first shallow angle relative to a tangent of the perimeter of the moderator such that a widest cross-section of the neutron-reflecting material of the reflector faces the set of fuel elements and reflects a first proportion of neutrons back into the set of fuel elements to increase a frequency of neutron-neutron collisions and increase a rate of fission within the set of fuel elements; and

30

wherein, in the second position, the reflector:
 occupies a second wide angle, greater than the first shallow angle, relative to the tangent of the perimeter of the moderator such that a narrowest cross-section of the neutron-reflecting material of the reflector faces the set of fuel elements; and
 reflects a second proportion of neutrons less than the first proportion of neutrons back into the set of fuel elements to reduce a frequency of neutron-neutron collisions and reduce a rate of fission within the set of fuel elements.

18. The system of claim 16, further comprising a propellant reservoir:
 arranged upstream of the pump;
 interposed between the set of fuel elements and a payload; and
 configured to store a volume of the propellant comprising hydrogen, the volume of the propellant absorbing neutron emitted axially by the set of fuel elements during a fission reaction to shield the payload from radiation.

19. A system comprising:
 a pump configured to pump a propellant;
 a set of fuel elements, each fuel element of the set of fuel elements comprising:
 a set of low enriched uranium fuel particles defining a set of interstitial volumes through which the pump pumps the propellant;
 a moderator:
 defining a set of fuel bores, each fuel bore in the set of fuel bores occupied by a fuel element; and
 defining a set of moderator coolant channels, each moderator coolant channel:
 fluidly isolated from the set of fuel bores; and
 configured to pass propellant through the moderator to cool the moderator;
 a reflector arranged on a perimeter of the moderator:
 comprising a neutron-reflecting material; and
 operable in:
 an open position to release neutrons to an external environment; and
 a closed position to reflect incident neutrons back into the set of fuel elements; and
 a thrust nozzle:
 configured to outlet propellant to produce thrust; and
 comprising a nozzle coolant channel arranged within a wall of the thrust nozzle configured to exchange thermal energy with propellant pumped through the nozzle coolant channel.

20. The system of claim 19, wherein:
 each fuel bore is configured to direct propellant axially through the moderator; and
 each fuel element of the set of fuel elements defines a cylindrical geometry of concentric layers comprising:
 a cold shell defining a first array of perforations configured to direct the propellant radially through the cold shell;
 a fuel bed containing the set of low enriched uranium fuel particles;
 a hot shell defining a second array of perforations configured to direct the propellant radially through the hot shell; and
 an interior chamber configured to direct propellant axially out of the fuel element toward the nozzle.

* * * * *